COMPLETE LEARNING SESSION

Geography 03 - Vulcanism and Earthquakes / India Seismic Zones - Learner-v2 Refreshed

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

CONTENTS / SESSION INDEX

Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

Geography 03 - Vulcanism and Earthquakes / India Seismic Zones

Complete uncompressed learner-v2 session | GS-I + Prelims | Generated 23 August 2026 | Approval pending

> **Syllabus ownership:** Important geophysical phenomena-earthquakes, tsunami and volcanic activity-and the salient features of world and Indian physical geography. **Source order:** Geography Markdown owners and section-control files -> OCR-searchable standard geography and routed local UPSC papers/keys -> dated official current sources -> no Qdrant dependency. **Fact discipline: [FACT]** means directly supported by a repository, book, official paper/key or official live source. **[INFERENCE]** means analytical synthesis. **[LIMIT]** marks uncertainty, scope or a classification caution. **Ownership boundary:** Geography explains the physical process, spatial pattern, site response and risk geography. Detailed incident-command and relief administration remain Disaster-Management territory.

Package component	Count / status
Basic/core numbered sessions	19
Verified primary-owner PYQs	7
Cross-owned but directly relevant Mains PYQ	1
Original hard MCQs	40
Remedial MCQs	8
Original solved Mains practice	6
Objective-key policy	Stable topic seed; balanced, deterministic and non-patterned
2026 PYQ key status	Local provisional key only; not represented as final UPSC key

BASIC LEARNING SESSION

Refreshed teaching navigation

Distinct embedded teaching-navigation image. The separate continuous Cārvāka-style flowchart package remains an independent at-a-glance artifact. The master route is:

```
ENDOGENIC ENERGY
```

```
-> plate setting
-> stress, strain and faults
-> earthquake source and seismic waves
-> measurement, distribution and cascading hazards
-> magma properties, volcanic forms and eruption styles
-> boundary/hot-spot distribution and environmental effects
-> India's tectonic regions and seismic zonation
-> microzonation, resilient construction and community preparedness
```

Static/current firewall: a recent earthquake or eruption illustrates a known mechanism; it does not prove a deterministic prediction, validate a metaphysical claim, or automatically establish a new hazard zone.

SESSION 1 - ENDOGENIC PROCESSES AND THE PLATE-TECTONIC FRAME

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Endogenic processes are tectonic and magmatic processes driven by energy within Earth that deform the crust and build relief.

Technical definition: Plate tectonics explains endogenic belts through interactions among lithospheric plates at convergent, divergent and transform margins.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Plate tectonics explains why earthquakes, volcanism and mountain building cluster in narrow belts rather than occurring randomly.

MUST-WRITE KEYWORDS

- **endogenic processes**
- **lithospheric plates**
- **convergent margin**
- **divergent margin**
- **transform margin**
- **subduction**
- **earthquake belt**
- **tectonic belt**

How to use them: Frame the answer through endogenic processes; define lithospheric plates, connect convergent margin with divergent margin to explain the mechanism, and use transform margin for the decisive comparison or qualification.

Visual first - where the two hazards fit

```
Residual planetary heat + radioactive decay + gravity
                         |
                         v
                MANTLE CONVECTION / FLOW
                         |
                         v
            LITHOSPHERIC PLATES MOVE RELATIVELY
              /              |               \
             /               |                \
     CONVERGENT          DIVERGENT          TRANSFORM
     compression         tension            shear
   trench + arc +      ridge + rift      strike-slip fault
   deep earthquakes    shallow quakes      shallow quakes
        |
        +---- subduction fluids -> melting -> explosive arc volcanism

HOT SPOT / PLUME: volcanism can also occur within a plate.
```

[FACT] Plate tectonics explains why most strong earthquakes and active volcanoes form narrow belts rather than a random global scatter. Convergent margins generate the widest depth range of earthquakes. Divergent and transform margins mainly generate shallow earthquakes.

[FACT] Earthquakes need not be volcanic. The Himalaya is intensely seismic because of continent-continent shortening, yet it lacks the continuous oceanic-slab arc volcanism typical of the Andes or Japan.

[INFERENCE] The best explanatory chain is **internal energy -> plate movement -> stress or melting -> event -> landform/hazard**. This prevents the common error of treating "plate tectonics" as the immediate cause of every tremor without explaining rupture.

Boundary	Plate motion	Dominant earthquake pattern	Typical volcanism
Ocean-continent convergence	Oceanic plate subducts	Shallow to deep, often powerful	Continental volcanic arc; commonly explosive
Ocean-ocean convergence	One oceanic plate subducts	Shallow to deep	Island arc
Continent-continent convergence	Crust shortens and thickens	Mainly shallow/intermediate thrust seismicity	Limited arc volcanism after ocean closure
Divergence	Plates separate	Shallow normal-fault earthquakes	Basaltic ridge/rift volcanism
Transform	Plates slide past	Shallow strike-slip earthquakes	Usually little magma generation
Hot spot	Plate crosses thermal anomaly	Local volcanic/seismic activity	Intraplate shields or flood basalts

> **ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM:** Plate tectonics gives the spatial grammar of seismicity and vulcanism, but the examiner still expects the immediate mechanics—fault slip for an earthquake and magma ascent for an eruption.

CLOSING RECALL FLOW - ENDOGENIC PROCESSES AND THE PLATE-TECTONIC FRAME

```
START / CONCEPT: ENDOGENIC PROCESSES AND THE PLATE-TECTONIC FRAME
        |
        v
EXACT TERMS: endogenic processes · lithospheric plates · convergent margin · divergent margin ·
transform margin · subduction · earthquake belt · tectonic belt
        |
        v
MECHANISM / ARGUMENT: Mantle heat and gravity drive plate motion, while collision, subduction,
rifting and shearing concentrate crustal deformation at plate margins.
        |
        v
CONSEQUENCE / CONTRAST: Different boundary mechanics produce distinct belts of earthquakes,
volcanoes, trenches, ridges and fold mountains.
        |
        v
UPSC TRAP / ANSWER-USE: Do not treat every convergent belt as volcanically identical: continent-
continent collision can be highly seismic without a continuous subduction-arc volcano chain.
        |
        v
ANSWER-GRABBING FORMULATION: Plate tectonics explains why earthquakes, volcanism and mountain
building cluster in narrow belts rather than occurring randomly.
```

SESSION 2 - STRESS, STRAIN, ELASTIC REBOUND AND FAULTS

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Stress is force per unit area; strain is the deformation produced.

Technical definition: Elastic strain is recoverable until failure, whereas permanent deformation remains.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Elastic rebound converts long-term tectonic loading into short-duration ground shaking, explaining how gradual plate motion produces a sudden disaster trigger.

MUST-WRITE KEYWORDS

- **Stress**
- **Strain**
- **Elastic Rebound**
- **Faults**
- **Mnemonic**
- **N-T-D**

How to use them: Frame the answer through Stress; define Strain, connect Elastic Rebound with Faults to explain the mechanism, and use Mnemonic for the decisive comparison or qualification.

Visual first - stress inside a crustal block

```
COMPRESSION                    TENSION                        SHEAR
 --->  <---                    <---  --->                   ↑      ↓
  rock shortens                 rock stretches              blocks slide
      |                              |                          |
 reverse / thrust fault          normal fault              strike-slip fault

Locked fault through time:
unaltered -> bent/strained -> rupture starts -> slip propagates -> rebound + aftershocks
```

Fault type	Stress	Relative movement	Common setting	Exam-safe sketch clue
Normal	Tension	Hanging wall moves down	Rift/divergent setting	Crust lengthens
Reverse	Compression	Hanging wall moves up	Convergent setting	Crust shortens
Thrust	Compression	Low-angle reverse fault	Fold-thrust belt	Large horizontal shortening
Strike-slip	Shear	Horizontal displacement	Transform/intraplate fault	Offset stream/road
Oblique-slip	Combined	Vertical + horizontal	Complex boundaries	Two components

[FACT] Stress is force per unit area; strain is the deformation produced. Elastic strain is recoverable until failure, whereas permanent deformation remains.

[FACT] A fault is not simply a crack. It is a fracture or fracture zone across which displacement has occurred. A joint has no demonstrable displacement.

[INFERENCE] Fault maps identify where rupture has occurred or may be geologically possible; they do not by themselves provide a date-specific prediction. Activity, slip rate, palaeoseismic evidence, geodesy and instrumental seismicity matter.

[FACT] The first large rupture in a sequence is called the mainshock only retrospectively if no later event is larger. Smaller preceding events are foreshocks only after the mainshock is identified; aftershocks follow stress redistribution.

> **Mnemonic: N-T-D** = Normal-Tension-Down; **R-C-Up** = Reverse-Compression-Up.

> **ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM:** Elastic rebound converts long-term tectonic loading into short-duration ground shaking, explaining how gradual plate motion produces a sudden disaster trigger.

CLOSING RECALL FLOW - STRESS, STRAIN, ELASTIC REBOUND AND FAULTS

```
START / CONCEPT: STRESS, STRAIN, ELASTIC REBOUND AND FAULTS
        |
        v
EXACT TERMS: Stress · Strain · Elastic Rebound · Faults · Mnemonic · N-T-D
        |
        v
MECHANISM / ARGUMENT: Fault maps identify where rupture has occurred or may be geologically
possible; they do not by themselves provide a date-specific prediction.
        |
        v
CONSEQUENCE / CONTRAST: Elastic strain is recoverable until failure, whereas permanent deformation
remains.
        |
        v
UPSC TRAP / ANSWER-USE: Do not miss this limiting distinction: stress is force per unit area.
        |
        v
ANSWER-GRABBING FORMULATION: Elastic rebound converts long-term tectonic loading into short-
duration ground shaking, explaining how gradual plate motion produces a sudden disaster trigger.
```

SESSION 3 - EARTHQUAKE TERMINOLOGY, CAUSES AND CLASSIFICATIONS

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Most damaging earthquakes are shallow because the source is closer to the surface, but damage is not fixed by depth alone.

Technical definition: Intermediate and deep earthquakes are strongly associated with descending slabs in subduction zones.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Most damaging earthquakes are shallow because the source is closer to the surface, but damage is not fixed by depth alone.

MUST-WRITE KEYWORDS

- **Earthquake Terminology**
- **Causes**
- **Classifications**
- **Cause**
- **Tectonic, volcanic, collapse, explosion, induced/reservoir-associated**
- **Immediate source process**

How to use them: Frame the answer through Earthquake Terminology; define Causes, connect Classifications with Cause to explain the mechanism, and use Tectonic, volcanic, collapse, explosion, induced/reservoir-associated for the decisive comparison or qualification.

Classification matrix

Basis	Classes	What it tells the examiner
Cause	Tectonic, volcanic, collapse, explosion, induced/reservoir-associated	Immediate source process
Focal depth	Shallow 0-70 km; intermediate 70-300 km; deep 300-700 km	Likely plate setting and surface-effect pattern
Tectonic setting	Interplate, intraplate	Boundary versus plate-interior source
Sequence	Foreshock, mainshock, aftershock; swarm	Temporal relationship, not a magnitude scale
Wave source	Natural or anthropogenic	Needed for monitoring and discrimination

[FACT] Most damaging earthquakes are shallow because the source is closer to the surface, but damage is not fixed by depth alone. Magnitude, distance, rupture direction, soil and construction remain decisive.

[FACT] Intermediate and deep earthquakes are strongly associated with descending slabs in subduction zones. Deep-focus events are absent from ordinary transform faults and mid-ocean ridges.

[FACT] Reservoir-associated seismicity is linked to changes in pore pressure and loading near existing fractures; Koyna is India's standard example. It does not mean every reservoir causes a damaging earthquake.

[LIMIT] “Human-induced” does not mean humans create tectonic stress from nothing. Injection, extraction, mining or impoundment can alter stresses or pore pressure enough to trigger slip on a susceptible fault.

```
CAUSE TREE
Earthquake
+- Tectonic: sudden fault slip
+- Volcanic: magma movement / pressure change
+- Collapse: cave or mine-roof failure, usually local
+- Explosion: chemical or nuclear source
+- Induced: injection, extraction, mining, reservoir loading
```

> **ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM:** Focal depth is a source descriptor, whereas disaster severity is a relational outcome produced by source, path, site, exposure and vulnerability.

CLOSING RECALL FLOW - EARTHQUAKE TERMINOLOGY, CAUSES AND CLASSIFICATIONS

```
START / CONCEPT: EARTHQUAKE TERMINOLOGY, CAUSES AND CLASSIFICATIONS
        |
        v
EXACT TERMS: Earthquake Terminology · Causes · Classifications · Cause · Tectonic, volcanic,
collapse, explosion, induced/reservoir-associated · Immediate source process
        |
        v
MECHANISM / ARGUMENT: It does not mean every reservoir causes a damaging earthquake.
        |
        v
CONSEQUENCE / CONTRAST: The resulting consequence is that most damaging earthquakes are shallow
because the source is closer to the surface.
        |
        v
UPSC TRAP / ANSWER-USE: “Human-induced” does not mean humans create tectonic stress from nothing.
        |
        v
ANSWER-GRABBING FORMULATION: Most damaging earthquakes are shallow because the source is closer to
the surface, but damage is not fixed by depth alone.
```

SESSION 4 - FOCUS, EPICENTRE, SEISMOGRAPHS AND LOCATING AN EVENT

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: The “focus” is simplified as a point for location.

Technical definition: Modern location uses many stations, a velocity model and iterative computation.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> The “focus” is simplified as a point for location.

MUST-WRITE KEYWORDS

- **Focus**
- **Epicentre**
- **Seismographs**
- **Locating An Event**
- **Modern**
- **Three-circle triangulation**

How to use them: Frame the answer through Focus; define Epicentre, connect Seismographs with Locating An Event to explain the mechanism, and use Modern for the decisive comparison or qualification.

Focus-epicentre cross-section

```
                         Earth's surface
       Station A ●          EPICENTRE               ● Station B
                    \           |                 /
                     \          | depth          /
                      \         |               /
                       \        X FOCUS        /
                        \______/ \__________/
                           rupturing fault

Waves radiate from the rupture; the hypocentre is an estimated source point,
while a large earthquake actually ruptures a finite fault area.
```

Reading a basic seismogram

```
ground
motion        P arrives       S arrives               surface waves
  ^             /\              /\/\                /\/\/\/\/\/\/\
  |____________/  \______________/    \________________/              \____ time
                  <--- S - P ---> gives station-to-epicentre distance
```

[FACT] One station's P-S time difference estimates distance from the source but not direction. A circle can be drawn around that station. The intersection of circles from at least three stations gives an epicentral estimate.

[FACT] Modern location uses many stations, a velocity model and iterative computation. Three-circle triangulation is the learner model, not the full operational method.

[FACT] A seismic trace records amplitude against time. The earliest small motion is normally the P arrival, followed by S and then longer-period surface waves.

[LIMIT] The “focus” is simplified as a point for location. Large ruptures propagate over a plane, so finite-fault geometry and directivity matter for real shaking.

> **ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM:** Epicentral location is a travel-time inference: arrival differences yield distance, multiple stations yield position, and additional modelling estimates depth and rupture geometry.

CLOSING RECALL FLOW - FOCUS, EPICENTRE, SEISMOGRAPHS AND LOCATING AN EVENT

```
START / CONCEPT: FOCUS, EPICENTRE, SEISMOGRAPHS AND LOCATING AN EVENT
        |
        v
EXACT TERMS: Focus · Epicentre · Seismographs · Locating An Event · Modern · Three-circle
triangulation
        |
        v
MECHANISM / ARGUMENT: The earliest small motion is normally the P arrival, followed by S and then
longer-period surface waves.
        |
        v
CONSEQUENCE / CONTRAST: Three-circle triangulation is the learner model, not the full operational
method.
        |
        v
UPSC TRAP / ANSWER-USE: Do not miss this limiting distinction: the “focus” is simplified as a point
for location.
        |
        v
ANSWER-GRABBING FORMULATION: The “focus” is simplified as a point for location.
```

SESSION 5 - P, S, LOVE AND RAYLEIGH WAVES

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Arrival order is P first, S second and surface waves later; medium control is P through solids and liquids but S through solids only.

Technical definition: “Surface waves are most destructive” is a useful rule, not an absolute law.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Arrival order is P first, S second and surface waves later; medium control is P through solids and liquids but S through solids only.

MUST-WRITE KEYWORDS

- **P**
- **S**
- **Love**
- **Rayleigh Waves**
- **To-and-fro parallel to travel direction**
- **Solids and liquids; also gases in principle**

How to use them: Frame the answer through P; define S, connect Love with Rayleigh Waves to explain the mechanism, and use To-and-fro parallel to travel direction for the decisive comparison or qualification.

Wave	Particle motion	Medium	Relative speed	Main exam use
P	To-and-fro parallel to travel direction	Solids and liquids; also gases in principle	Fastest	First arrival; compression/dilation
S	Perpendicular to travel direction	Solids only	Slower than P	Liquid outer-core evidence
Love	Horizontal side-to-side shear	Surface layers	Usually faster surface mode	Strong horizontal structural stress
Rayleigh	Retrograde/elliptical rolling	Surface layers	Slow surface mode	Vertical + horizontal ground motion

```
P WAVE:  ->  ||||||  ->     particles oscillate <----> along travel

S WAVE:  ->  ~~~~~~  ->     particles oscillate ↑↓ across travel

LOVE:    surface shifts <----> horizontally

RAYLEIGH: surface particles roll in ellipses, like a water-wave motion
```

[FACT] UPSC Prelims 2023 tested two precise statements: P waves arrive before S waves, and their particle motions differ. Both statements are correct; the local official key was unavailable, so the workbook labels the answer inferred with high confidence.

[FACT] S waves require shear rigidity and therefore do not pass through a liquid. P waves can pass through both solid and liquid but change velocity and direction at material boundaries.

[INFERENCE] “Surface waves are most destructive” is a useful rule, not an absolute law. Near-fault pulses, high-frequency body waves, resonance and local construction can alter the dominant damage source.

> **ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM:** Arrival order is P first, S second and surface waves later; medium control is P through solids and liquids but S through solids only.

CLOSING RECALL FLOW - P, S, LOVE AND RAYLEIGH WAVES

```
START / CONCEPT: P, S, LOVE AND RAYLEIGH WAVES
        |
        v
EXACT TERMS: P · S · Love · Rayleigh Waves · To-and-fro parallel to travel direction · Solids and
liquids; also gases in principle
        |
        v
MECHANISM / ARGUMENT: S waves require shear rigidity and therefore do not pass through a liquid.
        |
        v
CONSEQUENCE / CONTRAST: P waves can pass through both solid and liquid but change velocity and
direction at material boundaries.
        |
        v
UPSC TRAP / ANSWER-USE: “Surface waves are most destructive” is a useful rule, not an absolute law.
        |
        v
ANSWER-GRABBING FORMULATION: Arrival order is P first, S second and surface waves later; medium
control is P through solids and liquids but S through solids only.
```

SESSION 6 - REFRACTION, DISCONTINUITIES AND SHADOW ZONES

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: The larger S-wave shadow is a state-of-matter test, while the P-wave shadow is principally a refraction geometry.

Technical definition: The reappearance of P phases beyond the shadow zone results from strong refraction through the core.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> The larger S-wave shadow is a state-of-matter test, while the P-wave shadow is principally a refraction geometry.

MUST-WRITE KEYWORDS

- **Refraction**
- **Discontinuities**
- **Shadow Zones**
- **If Earth**
- **Direct S waves**
- **Direct**

How to use them: Frame the answer through Refraction; define Discontinuities, connect Shadow Zones with If Earth to explain the mechanism, and use Direct S waves for the decisive comparison or qualification.

```
                EARTHQUAKE •
                   \  P and S
                    \        MANTLE
                     \       _______
                      \     /       \
                       \   / LIQUID  \
                        \ / OUTER     \
                        / \ CORE       /
     S waves stop ---- X   \          /
     P waves bend ----------\_______/

Approximate angular pattern from the epicentre:
0° ---------------- 103° | 103° -------- 142° | 142° -------- 180°
P + S commonly recorded   P direct shadow       P returns after refraction
S-wave direct shadow extends beyond about 103°
```

[FACT] If Earth were homogeneous, seismic rays would follow simpler paths. Observed velocity changes, reflections and refractions identify internal discontinuities.

[FACT] The approximate P-wave shadow extends from about 103° to 142° from the epicentre. Direct S waves are absent beyond about 103° because the outer core is liquid.

[FACT] The reappearance of P phases beyond the shadow zone results from strong refraction through the core. This is not evidence that the whole core is liquid; other wave evidence supports a solid inner core within the liquid outer core.

[LIMIT] Angles are approximate and phase-specific. Do not turn a classroom shadow-zone sketch into an exact operational travel-time model.

> **ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM:** The larger S-wave shadow is a state-of-matter test, while the P-wave shadow is principally a refraction geometry.

CLOSING RECALL FLOW - REFRACTION, DISCONTINUITIES AND SHADOW ZONES

```
START / CONCEPT: REFRACTION, DISCONTINUITIES AND SHADOW ZONES
        |
        v
EXACT TERMS: Refraction · Discontinuities · Shadow Zones · If Earth · Direct S waves · Direct
        |
        v
MECHANISM / ARGUMENT: The reappearance of P phases beyond the shadow zone results from strong
refraction through the core.
        |
        v
CONSEQUENCE / CONTRAST: This is not evidence that the whole core is liquid; other wave evidence
supports a solid inner core within the liquid outer core.
        |
        v
UPSC TRAP / ANSWER-USE: Do not turn a classroom shadow-zone sketch into an exact operational travel-
time model.
        |
        v
ANSWER-GRABBING FORMULATION: The larger S-wave shadow is a state-of-matter test, while the P-wave
shadow is principally a refraction geometry.
```

SESSION 7 - MAGNITUDE, INTENSITY AND LOGARITHMIC ENERGY

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: “Richter scale” properly refers to local magnitude developed for a particular instrumental and regional context.

Technical definition: Intensity is not a synonym for economic loss because two buildings at the same intensity can perform differently.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Magnitude measures the earthquake; intensity measures the earthquake-at-a-place.

MUST-WRITE KEYWORDS

- **Magnitude**
- **Intensity**
- **Logarithmic Energy**
- **Governing question**
- **Number per event**
- **One preferred estimate, subject to revision**

How to use them: Frame the answer through Magnitude; define Intensity, connect Logarithmic Energy with Governing question to explain the mechanism, and use Number per event for the decisive comparison or qualification.

Feature	Magnitude	Intensity
Governing question	How large was the source?	How strong were effects here?

Feature	Magnitude	Intensity
Number per event	One preferred estimate, subject to revision	Many observations/maps
Basis	Instrumental waveform/source parameters	Felt reports, damage and observed effects
Modern large-event language	Mw	MMI/MSK or another intensity scale
Main control	Rupture area, slip and rigidity	Magnitude + depth + distance + site + building

```
LOCAL-MAGNITUDE INTUITION
M 5 -> baseline
M 6 -> 10 times recorded amplitude; about 31.6 times energy
M 7 -> 100 times amplitude; about 1,000 times energy relative to M 5

This intuition does NOT convert magnitude directly into casualties or damage.
```

[FACT] “Richter scale” properly refers to local magnitude developed for a particular instrumental and regional context. It can saturate for very large earthquakes; Mw is preferred for large modern events.

[FACT] Seismic moment is proportional to rock rigidity × ruptured fault area × average slip. Mw converts this physical source measure to a logarithmic magnitude.

[INFERENCE] A moderate shallow event beneath soft alluvium and vulnerable masonry can cause worse loss than a larger deep event under well-engineered settlements.

[LIMIT] Magnitude estimates can be revised as more data arrive. Intensity is not a synonym for economic loss because two buildings at the same intensity can perform differently.

> **ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM:** Magnitude measures the earthquake; intensity measures the earthquake-at-a-place.

CLOSING RECALL FLOW - MAGNITUDE, INTENSITY AND LOGARITHMIC ENERGY

```
START / CONCEPT: MAGNITUDE, INTENSITY AND LOGARITHMIC ENERGY
        |
        v
EXACT TERMS: Magnitude · Intensity · Logarithmic Energy · Governing question · Number per event ·
One preferred estimate, subject to revision
        |
        v
MECHANISM / ARGUMENT: Mw converts this physical source measure to a logarithmic magnitude.
        |
        v
CONSEQUENCE / CONTRAST: Intensity is not a synonym for economic loss because two buildings at the
same intensity can perform differently.
        |
        v
UPSC TRAP / ANSWER-USE: Do not miss this limiting distinction: “Richter scale” properly refers to
local magnitude developed for a particular instrumental and regional...
        |
        v
ANSWER-GRABBING FORMULATION: Magnitude measures the earthquake; intensity measures the earthquake-
at-a-place.
```

SESSION 8 - GLOBAL EARTHQUAKE DISTRIBUTION AND PLATE MARGINS

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: The Circum-Pacific belt is the dominant global concentration of subduction-related earthquakes and volcanoes.

Technical definition: Ridge earthquakes are shallow because the lithosphere is thin and hot.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Depth distribution is the diagnostic map layer: only a descending slab generates a systematic inclined belt from shallow trench earthquakes to intermediate and deep events.

MUST-WRITE KEYWORDS

- **Global Earthquake Distribution**
- **Plate Margins**
- **The Circum-Pacific belt**
- **The Circum-Pacific**
- **The Mediterranean**
- **Himalayan**

How to use them: Frame the answer through Global Earthquake Distribution; define Plate Margins, connect The Circum-Pacific belt with The Circum-Pacific to explain the mechanism, and use The Mediterranean for the decisive comparison or qualification.

```
WORLD SEISMIC BELTS - SCHEMATIC, NOT TO SCALE

Pacific rim:
Andes -> Central America -> Cascades/Alaska -> Aleutians -> Japan
      -> Philippines -> Indonesia -> New Zealand
      = trenches + arcs + shallow/deep earthquakes + tsunami potential

Mediterranean-Himalayan:
Mediterranean -> Türkiye/Iran -> Hindu Kush -> Himalaya -> Myanmar
      = collision, subduction remnants, transform and complex continental deformation

Mid-ocean ridges:
Mid-Atlantic + Indian Ocean ridge system
      = divergence, basaltic volcanism and shallow earthquakes
```

[FACT] The Circum-Pacific belt is the dominant global concentration of subduction-related earthquakes and volcanoes. Avoid memorising an undated percentage as a timeless constant.

[FACT] The Mediterranean-Himalayan belt includes complex convergence from southern Europe through West and Central Asia to the Himalayan and Myanmar regions.

[FACT] Ridge earthquakes are shallow because the lithosphere is thin and hot. Transform offsets between ridge segments also produce shallow strike-slip events.

[FACT] Intraplate earthquakes occur away from active boundaries, commonly by reactivation of inherited faults or rifts. Bhuj, Latur and New Madrid show that plate interiors are relatively-not absolutely-stable.

[CA FACT - USGS, 7 June 2026] A reviewed Mw 7.8 earthquake occurred 25 km southwest of Kablalan, Philippines, at about 57 km depth. [INFERENCE] Its location reinforces the active western-Pacific plate-boundary belt; it does not prove that every Pacific earthquake produces a tsunami.

> **ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM:** Depth distribution is the diagnostic map layer: only a descending slab generates a systematic inclined belt from shallow trench earthquakes to intermediate and deep events.

CLOSING RECALL FLOW - GLOBAL EARTHQUAKE DISTRIBUTION AND PLATE MARGINS

```
START / CONCEPT: GLOBAL EARTHQUAKE DISTRIBUTION AND PLATE MARGINS
        |
        v
EXACT TERMS: Global Earthquake Distribution · Plate Margins · The Circum-Pacific belt · The Circum-
Pacific · The Mediterranean · Himalayan
        |
        v
MECHANISM / ARGUMENT: Ridge earthquakes are shallow because the lithosphere is thin and hot.
        |
        v
CONSEQUENCE / CONTRAST: Bhuj, Latur and New Madrid show that plate interiors are relatively-not
absolutely-stable.
        |
        v
UPSC TRAP / ANSWER-USE: Do not miss this limiting distinction: the Circum-Pacific belt is the
dominant global concentration of subduction-related earthquakes and volcanoes.
        |
        v
ANSWER-GRABBING FORMULATION: Depth distribution is the diagnostic map layer: only a descending slab
generates a systematic inclined belt from shallow trench earthquakes to intermediate and deep
events.
```

SESSION 9 - SECONDARY HAZARDS, SITE RESPONSE AND THE RISK EQUATION

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Hazard is physical possibility; vulnerability is susceptibility; risk is their interaction with exposure; disaster is realised severe disruption beyond coping capacity.

Technical definition: Urban disaster risk is often a network problem: a structurally surviving hospital may still fail functionally if power, water, access or communications collapse.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Hazard is physical possibility; vulnerability is susceptibility; risk is their interaction with exposure; disaster is realised severe disruption beyond coping capacity.

MUST-WRITE KEYWORDS

- **Secondary Hazards**
- **Site Response**
- **The Risk Equation**
- **Liquefaction**
- **Floodplains, deltas, reclaimed land, old channels**
- **Lateral spreading**

How to use them: Frame the answer through Secondary Hazards; define Site Response, connect The Risk Equation with Liquefaction to explain the mechanism, and use Floodplains, deltas, reclaimed land, old channels for the decisive comparison or qualification.

```
SEISMIC RISK CHAIN
Fault rupture
   -> shaking + surface displacement
   -> local ground response
      +- soft-soil amplification
      +- liquefaction / lateral spreading
      +- slope failure
   -> built-system cascades
      +- fire from broken utilities
      +- bridge/road/port failure
      +- dam or embankment distress
      +- hospital/communication disruption
   -> loss depends on EXPOSURE × VULNERABILITY ÷ CAPACITY
```

Hazard	Mechanism	Spatial clue
Liquefaction	Shaking raises pore pressure in loose saturated sediment until effective strength falls	Floodplains, deltas, reclaimed land, old channels
Lateral spreading	Liquefied soil moves towards a free face or down gentle slope	Riverbanks, ports, embankments
Landslide/rockfall	Shaking destabilises marginal slopes	Himalaya, road cuts, steep valleys
Fire	Gas/electric lines break while access and water supply fail	Dense urban areas
Surface rupture	Fault displacement reaches ground	Narrow fault corridor
Dam/lifeline failure	Shaking, settlement or slope collapse damages critical systems	Reservoir valleys and urban networks

[FACT] Soft sediment can amplify selected frequencies as waves slow and grow in amplitude. Basin geometry may prolong shaking.

[FACT] Liquefaction is not “soil melting.” It is a temporary loss of effective stress and strength in susceptible saturated granular material.

[INFERENCE] Urban disaster risk is often a network problem: a structurally surviving hospital may still fail functionally if power, water, access or communications collapse.

> **ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM:** Hazard is physical possibility; vulnerability is susceptibility; risk is their interaction with exposure; disaster is realised severe disruption beyond coping capacity.

CLOSING RECALL FLOW - SECONDARY HAZARDS, SITE RESPONSE AND THE RISK EQUATION

```
START / CONCEPT: SECONDARY HAZARDS, SITE RESPONSE AND THE RISK EQUATION
        |
        v
EXACT TERMS: Secondary Hazards · Site Response · The Risk Equation · Liquefaction · Floodplains,
deltas, reclaimed land, old channels · Lateral spreading
        |
        v
MECHANISM / ARGUMENT: Urban disaster risk is often a network problem: a structurally surviving
hospital may still fail functionally if power, water, access or communications collapse.
        |
        v
CONSEQUENCE / CONTRAST: Liquefaction is not “soil melting.” It is a temporary loss of effective
stress and strength in susceptible saturated granular material.
        |
        v
UPSC TRAP / ANSWER-USE: Do not miss this limiting distinction: soft sediment can amplify selected
frequencies as waves slow and grow in amplitude.
        |
        v
ANSWER-GRABBING FORMULATION: Hazard is physical possibility; vulnerability is susceptibility; risk
is their interaction with exposure; disaster is realised severe disruption beyond coping capacity.
```

SESSION 10 - TSUNAMI GENERATION, PROPAGATION AND COASTAL IMPACT

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Strike-slip movement with little vertical displacement is less efficient at generating a large basin-wide tsunami than thrust displacement, although landslides triggered by any event may create a local tsunami.

Technical definition: In deep water, a tsunami can pass beneath ships with little notice.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Strike-slip movement with little vertical displacement is less efficient at generating a large basin-wide tsunami than thrust displacement, although landslides triggered by any event may create a local tsunami.

MUST-WRITE KEYWORDS

- **Tsunami Generation**
- **Propagation**
- **Coastal Impact**
- **Strike-slip movement with little vertical displacement**
- **Strike-slip**
- **In**

How to use them: Frame the answer through Tsunami Generation; define Propagation, connect Coastal Impact with Strike-slip movement with little vertical displacement to explain the mechanism, and use Strike-slip for the decisive comparison or qualification.

```
SUBDUCTION MEGATHRUST
oceanic plate dives -> locked interface deforms -> sudden rupture
                               |
                               v
                    sea floor moves vertically
                               |
                               v
               whole water column is displaced
                               |
          deep ocean: long, fast, low amplitude
                               |
                               v
             shallow shelf: speed falls, height rises
                               |
                               v
                  RUN-UP + INUNDATION + RETURN FLOW
```

[FACT] Strike-slip movement with little vertical displacement is less efficient at generating a large basin-wide tsunami than thrust displacement, although landslides triggered by any event may create a local tsunami.

[FACT] In deep water, a tsunami can pass beneath ships with little notice. Near shore, decreasing depth slows the wave, shortens wavelength and increases height.

[FACT] The first arrival need not be the largest, and sea withdrawal need not precede every tsunami. Strong or long coastal shaking, unusual sea-level change or an official alert requires immediate movement along identified evacuation routes.

[FACT] On 26 December 2004, rupture on the Sunda subduction margin generated the Indian Ocean tsunami. India subsequently established the Indian Tsunami Early Warning Centre at INCOIS, Hyderabad, using seismic, bottom-pressure and tide-gauge observations.

[INFERENCE] Coastal damage varies with bathymetry, shelf width, bays, estuaries, coastal slope, settlement pattern and evacuation capacity. “Move to 10 metres” is not a universal rule; follow official local maps and routes.

> **ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM:** Tsunami risk joins a tectonic source to an oceanographic transmission path and a geomorphically conditioned coast.

CLOSING RECALL FLOW - TSUNAMI GENERATION, PROPAGATION AND COASTAL IMPACT

```
START / CONCEPT: TSUNAMI GENERATION, PROPAGATION AND COASTAL IMPACT
        |
        v
EXACT TERMS: Tsunami Generation · Propagation · Coastal Impact · Strike-slip movement with little
vertical displacement · Strike-slip · In
        |
        v
MECHANISM / ARGUMENT: "Move to 10 metres" is not a universal rule; follow official local maps and
routes.
        |
        v
CONSEQUENCE / CONTRAST: In deep water, a tsunami can pass beneath ships with little notice.
        |
        v
UPSC TRAP / ANSWER-USE: Do not miss this limiting distinction: strong or long coastal shaking,
unusual sea-level change or an official alert requires immediate...
        |
        v
ANSWER-GRABBING FORMULATION: Strike-slip movement with little vertical displacement is less
efficient at generating a large basin-wide tsunami than thrust displacement, although landslides
triggered by any event may create a local tsunami.
```

SESSION 11 - VOLCANO ANATOMY, MAGMA, LAVA, SILICA, GAS AND VISCOSITY

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Basaltic magma is generally hotter, less silica-rich and less viscous than andesitic or rhyolitic magma.

Technical definition: Rhyolitic magma is commonly cooler and more viscous.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Silica raises viscosity, viscosity impedes degassing, retained gas raises pressure, and pressure-driven fragmentation makes an eruption explosive.

MUST-WRITE KEYWORDS

- **Volcano Anatomy**
- **Magma**
- **Lava**
- **Silica**
- **Gas**
- **Viscosity**

How to use them: Frame the answer through Volcano Anatomy; define Magma, connect Lava with Silica to explain the mechanism, and use Gas for the decisive comparison or qualification.

Volcano cross-section

```
                         ash / gas plume
                              ↑
                      ______ crater ______
                     /       central       \
       lava flow -->/          vent         \ <-- layers of lava + tephra
                   /            |            \
                  /       side vent -> cone   \
-----------------/--------------|-------------\------ surface
                                |
                             conduit
                    dyke  /     |       \  sill
                         /      |        \______
                        /     MAGMA CHAMBER     \
                       /________________________\

Caldera: broad collapse depression after major magma withdrawal; not simply a large crater.
```

Control	Effect on viscosity/degassing	Likely tendency
More silica	Greater polymerisation	Higher viscosity
Higher temperature	Easier molecular movement	Lower viscosity
More crystals	Stiffer magma	Higher effective viscosity
Dissolved gas retained under pressure	Potential rapid expansion during ascent	More explosive fragmentation
Open conduits/easy degassing	Pressure escapes	More effusive behaviour

[FACT] Basaltic magma is generally hotter, less silica-rich and less viscous than andesitic or rhyolitic magma. Rhyolitic magma is commonly cooler and more viscous.

[FACT] Pressure decreases during ascent. Dissolved volatiles exsolve into bubbles; if viscous magma prevents escape, gas pressure can fragment magma into tephra and drive explosive eruptions.

[LIMIT] Composition gives a tendency, not a guaranteed eruption. Water-magma interaction, conduit sealing, eruption rate and external water can change behaviour.

[FACT] A crater is the vent-centred depression at a summit or flank. A caldera is a much larger collapse structure commonly related to withdrawal of a substantial magma volume.

> **ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM:** Silica raises viscosity, viscosity impedes degassing, retained gas raises pressure, and pressure-driven fragmentation makes an eruption explosive.

CLOSING RECALL FLOW - VOLCANO ANATOMY, MAGMA, LAVA, SILICA, GAS AND VISCOSITY

```
START / CONCEPT: VOLCANO ANATOMY, MAGMA, LAVA, SILICA, GAS AND VISCOSITY
        |
        v
EXACT TERMS: Volcano Anatomy · Magma · Lava · Silica · Gas · Viscosity
        |
        v
MECHANISM / ARGUMENT: Dissolved volatiles exsolve into bubbles; if viscous magma prevents escape,
gas pressure can fragment magma into tephra and drive explosive eruptions.
        |
        v
CONSEQUENCE / CONTRAST: Basaltic magma is generally hotter, less silica-rich and less viscous than
andesitic or rhyolitic magma.
        |
        v
UPSC TRAP / ANSWER-USE: Do not miss this limiting distinction: water-magma interaction, conduit
sealing, eruption rate and external water can change behaviour.
        |
        v
ANSWER-GRABBING FORMULATION: Silica raises viscosity, viscosity impedes degassing, retained gas
raises pressure, and pressure-driven fragmentation makes an eruption explosive.
```

SESSION 12 - INTRUSIVE AND EXTRUSIVE FORMS; VOLCANIC PRODUCTS

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Intrusive And Extrusive Forms; Volcanic Products comprises Intrusive, Extrusive Forms and Volcanic Products as its core connected dimensions.

Technical definition: Product classification is also a risk map: ballistic blocks dominate near the vent, ash affects broad downwind areas, pyroclastic currents follow topography, and lahars extend far along drainage lines.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Intrusive And Extrusive Forms; Volcanic Products comprises Intrusive, Extrusive Forms and Volcanic Products as its core connected dimensions.

MUST-WRITE KEYWORDS

- **Intrusive**
- **Extrusive Forms**
- **Volcanic Products**
- **D: all four**
- **Lava flow**
- **Molten rock at surface**

How to use them: Frame the answer through Intrusive; define Extrusive Forms, connect Volcanic Products with D: all four to explain the mechanism, and use Lava flow for the decisive comparison or qualification.

Intrusive forms - labelled schematic

```
sedimentary layers  ================================
                    =======  SILL  ================  concordant sheet
                    =======________================
                       /  LACCOLITH  \              dome-shaped concordant
----------------------/--------------\---------------
              | DYKE |                              discordant sheet
              |      |      ( PHACOLITH )           fold-related lens
              |      |     ________________
              |      |    /    BATHOLITH    \       very large deep body
______________|______|___/___________________\_______
```

Product/form	Meaning	Major hazard or landform
Lava flow	Molten rock at surface	Buries/burns land; basalt can travel far
Volcanic gas	H_2O, CO_2, SO_2 and other species including sulphur and nitrogen species	Toxicity, acid aerosols, climate/air effects
Ash/dust	Fine pyroclasts	Roof load, crops, water, health, aviation
Lapilli	2-64 mm pyroclasts	Fallout and cone building
Bomb/block	Coarse fragment	Ballistic hazard near vent
Pyroclastic density current	Hot turbulent gas-particle flow	Fast, ground-hugging, lethal
Lahar	Water-mobilised volcanic debris	Valley-confined mudflow, can occur long after eruption
Fissure basalt	Fluid lava from cracks	Lava plateau, e.g. Deccan Traps

[FACT] UPSC Prelims 2024 asked whether pyroclastic debris, ash and dust, nitrogen compounds and sulphur compounds are products of volcanic eruptions. The official local Set-A key is **D: all four.**

[FACT] Dykes cut across pre-existing layers; sills run approximately parallel to them. Batholiths are very large discordant plutonic bodies later exposed by uplift and erosion.

[FACT] A lahar is not lava. It is a flowing mixture of water and volcanic sediment, often channelled down valleys.

[INFERENCE] Product classification is also a risk map: ballistic blocks dominate near the vent, ash affects broad downwind areas, pyroclastic currents follow topography, and lahars extend far along drainage lines.

> **ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM:** The same eruption has a spatially differentiated hazard field—ballistics near the vent, density currents on flanks, ash downwind and lahars down valleys.

CLOSING RECALL FLOW - INTRUSIVE AND EXTRUSIVE FORMS; VOLCANIC PRODUCTS

```
START / CONCEPT: INTRUSIVE AND EXTRUSIVE FORMS; VOLCANIC PRODUCTS
        |
        v
EXACT TERMS: Intrusive · Extrusive Forms · Volcanic Products · D: all four · Lava flow · Molten
rock at surface
        |
        v
MECHANISM / ARGUMENT: Batholiths are very large discordant plutonic bodies later exposed by uplift
and erosion.
        |
        v
CONSEQUENCE / CONTRAST: UPSC Prelims 2024 asked whether pyroclastic debris, ash and dust, nitrogen
compounds and sulphur compounds are products of volcanic eruptions.
        |
        v
UPSC TRAP / ANSWER-USE: Do not miss this limiting distinction: it is a flowing mixture of water and
volcanic sediment, often channelled down valleys.
        |
        v
ANSWER-GRABBING FORMULATION: Intrusive And Extrusive Forms; Volcanic Products comprises Intrusive,
Extrusive Forms and Volcanic Products as its core connected dimensions.
```

SESSION 13 - ERUPTION STYLES, VOLCANO TYPES AND ACTIVITY STATUS

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: “Active, dormant and extinct” are descriptive status classes, not a compulsory life cycle.

Technical definition: A volcano with no historical eruption may still be capable of renewed activity if geological and geophysical evidence shows a live system.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Active status concerns eruptive capability; shield/composite concerns form; Hawaiian/Plinian concerns behaviour-UPSC can mix these axes in close options.

MUST-WRITE KEYWORDS

- **Eruption Styles**
- **Volcano Types**
- **Activity Status**
- **Icelandic/fissure**
- **Quiet, voluminous basalt from long cracks**
- **Lava plateau**

How to use them: Frame the answer through Eruption Styles; define Volcano Types, connect Activity Status with Icelandic/fissure to explain the mechanism, and use Quiet, voluminous basalt from long cracks for the decisive comparison or qualification.

Eruption style	Core behaviour	Common product/shape
Icelandic/fissure	Quiet, voluminous basalt from long cracks	Lava plateau
Hawaiian	Fluid lava fountains and flows	Shield volcano
Strombolian	Repeated moderate bursts	Scoria/cinder cone
Vulcanian	Short violent ash-rich blasts from plugged vent	Ash and blocks
Plinian/Vesuvian	Sustained high eruption column, widespread tephra	Explosive stratovolcano/caldera potential
Pelean/dome-collapse	Viscous dome growth and collapse	Pyroclastic density currents

Volcano type	Profile	Magma/eruption tendency	Example route
Shield	Broad, gentle	Basaltic, effusive	Hawai‘i
Composite/strato	Tall, layered, steep upper slopes	Andesitic-dacitic, mixed/explosive	Andes, Japan
Cinder/scoria cone	Small, steep fragment cone	Strombolian fragments	Monogenetic fields
Lava dome	Steep viscous mound	Silicic, unstable collapse	Arc volcano summits
Caldera system	Broad collapse depression	Major withdrawal/explosive history	Crater-lake systems
Fissure plateau	Sheet lava, step-like relief	Fluid flood basalt	Deccan

[FACT] “Active, dormant and extinct” are descriptive status classes, not a compulsory life cycle. A volcano with no historical eruption may still be capable of renewed activity if geological and geophysical evidence shows a live system.

[FACT] Barren Island is India's active-volcano example within the Andaman volcanic arc. Narcondam is volcanic but should not be substituted for Barren Island in the 2018 PYQ.

[CA FACT - USGS/HVO, 10 March 2026] Kīlauea episode 43 produced vigorous lava fountains and extensive tephra fallout. [INFERENCE] It is a current illustration of basaltic volcanism producing both effusive lava and hazardous fragment fallout; “basaltic” does not mean harmless.

> **ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM:** Active status concerns eruptive capability; shield/composite concerns form; Hawaiian/Plinian concerns behaviour—UPSC can mix these axes in close options.

CLOSING RECALL FLOW - ERUPTION STYLES, VOLCANO TYPES AND ACTIVITY STATUS

```
START / CONCEPT: ERUPTION STYLES, VOLCANO TYPES AND ACTIVITY STATUS
        |
        v
EXACT TERMS: Eruption Styles · Volcano Types · Activity Status · Icelandic/fissure · Quiet,
voluminous basalt from long cracks · Lava plateau
        |
        v
MECHANISM / ARGUMENT: A volcano with no historical eruption may still be capable of renewed
activity if geological and geophysical evidence shows a live system.
        |
        v
CONSEQUENCE / CONTRAST: “Active, dormant and extinct” are descriptive status classes, not a
compulsory life cycle.
        |
        v
UPSC TRAP / ANSWER-USE: Narcondam is volcanic but should not be substituted for Barren Island in
the 2018 PYQ.
        |
        v
ANSWER-GRABBING FORMULATION: Active status concerns eruptive capability; shield/composite concerns
form; Hawaiian/Plinian concerns behaviour-UPSC can mix these axes in close options.
```

SESSION 14 - PLATE-SETTING VOLCANISM, RING OF FIRE AND MID-OCEAN RIDGES

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: The Ring of Fire is a linked trench-earthquake-arc system, not merely a ring-shaped list of volcanoes.

Technical definition: The Ring of Fire includes trenches, island arcs, continental arcs, transform segments, earthquakes from shallow to deep, explosive volcanoes and tsunami-generating megathrusts around much of the Pacific margin.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> The Ring of Fire is a linked trench-earthquake-arc system, not merely a ring-shaped list of volcanoes.

MUST-WRITE KEYWORDS

- **Plate-Setting Volcanism**
- **Ring Of Fire**
- **Mid-Ocean Ridges**
- **The Ring**
- **Fire**
- **Pacific**

How to use them: Frame the answer through Plate-Setting Volcanism; define Ring Of Fire, connect Mid-Ocean Ridges with The Ring to explain the mechanism, and use Fire for the decisive comparison or qualification.

```
SUBDUCTION MARGIN
oceanic plate ---> trench \ descends
                              \ water-rich minerals release volatiles
                               \        ↑
                                \  MANTLE WEDGE melts
                                 \       ↑ magma
continent / island arc ___________\______/\/\ volcanoes
            shallow quakes -> intermediate -> deep along slab

DIVERGENT MARGIN
plate <---      ridge/rift      ---> plate
                  ↑ hot mantle rises
                  ↑ pressure falls
             decompression melting -> basalt
```

[FACT] The Ring of Fire includes trenches, island arcs, continental arcs, transform segments, earthquakes from shallow to deep, explosive volcanoes and tsunami-generating megathrusts around much of the Pacific margin.

[FACT] Mid-ocean ridges form the longest volcanic system on Earth, mostly submarine. New basaltic crust forms at spreading axes.

[FACT] Continental rifts may begin with normal faulting and basaltic volcanism; continued extension can develop a narrow sea and eventually an ocean basin.

[FACT] The Mediterranean-Himalayan belt is not one uniform volcanic arc. It combines subduction remnants, collision, escape tectonics and complex faults; the Himalaya itself is primarily a collision-seismic belt.

```
RING OF FIRE ANSWER MAP LABELS
Andes | Central America | Cascades | Alaska-Aleutians | Kamchatka
Japan | Philippines | Indonesia | Tonga-Kermadec | New Zealand
```

> **ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM:** The Ring of Fire is a linked trench–earthquake–arc system, not merely a ring-shaped list of volcanoes.

CLOSING RECALL FLOW - PLATE-SETTING VOLCANISM, RING OF FIRE AND MID-OCEAN RIDGES

```
START / CONCEPT: PLATE-SETTING VOLCANISM, RING OF FIRE AND MID-OCEAN RIDGES
        |
        v
EXACT TERMS: Plate-Setting Volcanism · Ring Of Fire · Mid-Ocean Ridges · The Ring · Fire · Pacific
        |
        v
MECHANISM / ARGUMENT: The Ring of Fire includes trenches, island arcs, continental arcs, transform
segments, earthquakes from shallow to deep, explosive volcanoes and tsunami-generating megathrusts
around much of the Pacific margin.
        |
        v
CONSEQUENCE / CONTRAST: The Mediterranean-Himalayan belt is not one uniform volcanic arc.
        |
        v
UPSC TRAP / ANSWER-USE: Do not miss this limiting distinction: continental rifts may begin with
normal faulting and basaltic volcanism.
        |
        v
ANSWER-GRABBING FORMULATION: The Ring of Fire is a linked trench-earthquake-arc system, not merely
a ring-shaped list of volcanoes.
```

SESSION 15 - MANTLE PLUMES, HOT SPOTS AND FLOOD BASALTS

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Mantle Plumes, Hot Spots And Flood Basalts comprises Mantle Plumes, Hot Spots and Flood Basalts as its core connected dimensions.

Technical definition: Flood basalts erupt mainly from fissures and build plateaux through repeated fluid flows.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Deccan volcanism is associated in standard plume interpretation with the Réunion hot-spot system and the Indian Plate's passage, while the Hawaiian chain illustrates a long-lived oceanic hot spot.

MUST-WRITE KEYWORDS

- **Mantle Plumes**
- **Hot Spots**
- **Flood Basalts**
- **The classic hot-spot expectation**
- **Flood**
- **The Deccan Traps**

How to use them: Frame the answer through Mantle Plumes; define Hot Spots, connect Flood Basalts with The classic hot-spot expectation to explain the mechanism, and use Flood for the decisive comparison or qualification.

```
                 moving lithospheric plate  ----------->
surface       old volcano      older      young      ACTIVE
                 /\             /\         /\          /\
-----------------|--------------|----------|-----------|-------
                 |              |          |           |
                 +-------------- age increases --------+
                                                plume tail
                                                    ↑
                                              buoyant upwelling
                                             /                \
                                      earlier broad PLUME HEAD
                                      -> flood-basalt outpouring
```

[FACT] The classic hot-spot expectation is an age-progressive chain, with the active volcano over or near the present source and older extinct centres trailing in the direction opposite plate motion.

[FACT] Flood basalts erupt mainly from fissures and build plateaux through repeated fluid flows. The Deccan Traps are not a giant single cone.

[FACT] Deccan volcanism is associated in standard plume interpretation with the Réunion hot-spot system and the Indian Plate's passage, while the Hawaiian chain illustrates a long-lived oceanic hot spot.

[INFERENCE] Hot spots can supply a reference frame for plate direction and relative rate independent of ridge magnetic stripes, though plume motion and mantle flow complicate a perfectly fixed frame.

[LIMIT] The deep-plume hypothesis is influential but not the only explanation of intraplate volcanism; lithospheric extension, edge-driven convection and shallow mantle heterogeneity may also contribute.

> **ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM:** Plumes add a vertical mantle component to the horizontal plate-tectonic system, but they refine rather than replace plate tectonics.

CLOSING RECALL FLOW - MANTLE PLUMES, HOT SPOTS AND FLOOD BASALTS

```
START / CONCEPT: MANTLE PLUMES, HOT SPOTS AND FLOOD BASALTS
        |
        v
EXACT TERMS: Mantle Plumes · Hot Spots · Flood Basalts · The classic hot-spot expectation · Flood ·
The Deccan Traps
        |
        v
MECHANISM / ARGUMENT: Flood basalts erupt mainly from fissures and build plateaux through repeated
fluid flows.
        |
        v
CONSEQUENCE / CONTRAST: Hot spots can supply a reference frame for plate direction and relative
rate independent of ridge magnetic stripes, though plume motion and mantle flow complicate a
perfectly fixed frame.
        |
        v
UPSC TRAP / ANSWER-USE: The deep-plume hypothesis is influential but not the only explanation of
intraplate volcanism; lithospheric extension, edge-driven convection and shallow mantle
heterogeneity may also contribute.
        |
        v
ANSWER-GRABBING FORMULATION: Deccan volcanism is associated in standard plume interpretation with
the Réunion hot-spot system and the Indian Plate's passage, while the Hawaiian chain illustrates a
long-lived oceanic hot spot.
```

SESSION 16 - CONSTRUCTIVE, DESTRUCTIVE, CLIMATIC AND ENVIRONMENTAL EFFECTS

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: The climatic effect depends on what reaches the stratosphere, while the regional effect depends on where ash, lava, pyroclastic currents and lahars travel.

Technical definition: Technically, Constructive, Destructive, Climatic And Environmental Effects is analysed by relating Constructive to Destructive, then testing the relationship through Climatic and Environmental Effects.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> The climatic effect depends on what reaches the stratosphere, while the regional effect depends on where ash, lava, pyroclastic currents and lahars travel.

MUST-WRITE KEYWORDS

- **Constructive**
- **Destructive**
- **Climatic**
- **Environmental Effects**
- **Lava**
- **Burial, burning, cutting access**

How to use them: Frame the answer through Constructive; define Destructive, connect Climatic with Environmental Effects to explain the mechanism, and use Lava for the decisive comparison or qualification.

Pathway	Immediate process	Environmental/regional impact
Lava	Burial, burning, cutting access	Land/infrastructure loss; new rock and coast
Ash	Downwind fallout	Crop loss, roof loading, water contamination, aviation disruption
Density current	Hot gas-particle flow	Extreme proximal mortality and ecosystem stripping
Lahar	Water + ash/debris down valleys	Repeated valley destruction after eruption

Pathway	Immediate process	Environmental/regional impact
SO2 to stratosphere	Sulphate aerosol reflects sunlight	Temporary regional/hemispheric cooling
CO2 and other gases	Local accumulation or long-term greenhouse contribution	Local asphyxiation; different timescale from aerosol cooling
Weathering/hydrothermal action	Nutrient release and mineral deposition	Fertile soils, ore bodies, geothermal use

[FACT] Short-lived cooling after a major explosive eruption is principally associated with stratospheric sulphate aerosols, not with ash remaining aloft indefinitely.

[FACT] Ash can eventually contribute nutrients, but fresh ash may initially smother crops, contaminate water and damage machinery. “Volcanic soil is fertile” should never erase the transition cost.

[FACT] Volcanic landscapes support tourism, geothermal power and mineral resources, but these benefits may also increase settlement and exposure.

[FACT] The 2021 eruptions used in the Mains PYQ illustrate different pathways: La Soufrière ash and pyroclastic activity in St Vincent; Nyiragongo lava near Goma; Cumbre Vieja lava/ash on La Palma; Semeru pyroclastic and lahar hazards in Indonesia.

[INFERENCE] The answer should be regional, not merely global: wind, valleys, island size, farming systems, water supply and aviation routes determine who experiences which impact.

> **ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM:** The climatic effect depends on what reaches the stratosphere, while the regional effect depends on where ash, lava, pyroclastic currents and lahars travel.

CLOSING RECALL FLOW - CONSTRUCTIVE, DESTRUCTIVE, CLIMATIC AND ENVIRONMENTAL EFFECTS

```
START / CONCEPT: CONSTRUCTIVE, DESTRUCTIVE, CLIMATIC AND ENVIRONMENTAL EFFECTS
        |
        v
EXACT TERMS: Constructive · Destructive · Climatic · Environmental Effects · Lava · Burial, burning,
 cutting access
        |
        v
MECHANISM / ARGUMENT: Short-lived cooling after a major explosive eruption is principally
associated with stratospheric sulphate aerosols, not with ash remaining aloft indefinitely.
        |
        v
CONSEQUENCE / CONTRAST: Ash can eventually contribute nutrients, but fresh ash may initially
smother crops, contaminate water and damage machinery.
        |
        v
UPSC TRAP / ANSWER-USE: Volcanic landscapes support tourism, geothermal power and mineral resources,
 but these benefits may also increase settlement and exposure.
        |
        v
ANSWER-GRABBING FORMULATION: The climatic effect depends on what reaches the stratosphere, while
the regional effect depends on where ash, lava, pyroclastic currents and lahars travel.
```

SESSION 17 - INDIA'S TECTONIC FRAMEWORK AND REGIONAL SEISMIC RISK

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: India's earthquake geography must connect tectonic source, geomorphic site and settlement vulnerability; a zone label alone cannot explain loss.

Technical definition: The 2001 Bhuj earthquake demonstrates severe intraplate risk in Kachchh and the role of non-engineered construction and liquefaction.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> India's earthquake geography must connect tectonic source, geomorphic site and settlement vulnerability; a zone label alone cannot explain loss.

MUST-WRITE KEYWORDS

- India'S Tectonic Framework
- Regional Seismic Risk
- Himalaya
- Ongoing continent-continent shortening and thrusting
- Steep slopes, road isolation, vulnerable hill construction
- NE/Indo-Burmese

How to use them: Frame the answer through India'S Tectonic Framework; define Regional Seismic Risk, connect Himalaya with Ongoing continent-continent shortening and thrusting to explain the mechanism, and use Steep slopes, road isolation, vulnerable hill construction for the decisive comparison or qualification.

```
INDIA SEISMOTECTONIC SCHEMATIC - NOT A LEGAL BOUNDARY MAP

NW Himalaya / Kashmir ---- Central Himalaya ---- Eastern Himalaya
       collision, shortening, thrust loading; young relief and landslides
                                      \
                                       \  NE syntaxis + Indo-Burmese ranges
                                        \ complex compression/strike-slip

KACHCHH: inherited rift/fault reactivation        PENINSULAR SHIELD:
Bhuj 2001, intraplate high hazard                 relatively stable, not aseismic
                                                   Koyna / Latur examples

Bay of Bengal ------------------ ANDAMAN-NICOBAR ARC / SUNDA TRENCH
                                  subduction + earthquakes + Barren Island + tsunami
```

Region	Tectonic logic	Risk modifier / case
Himalaya	Ongoing continent-continent shortening and thrusting	Steep slopes, road isolation, vulnerable hill construction
NE/Indo-Burmese	Interaction of Himalayan and Burma/Sunda systems	Complex faults, high rainfall and landslide cascades
Andaman-Nicobar	Active subduction and island arc	Earthquake-tsunami-volcanic multi-hazard
Kachchh	Reactivated inherited rift/fault structures	Bhuj 2001; liquefaction and masonry vulnerability
Peninsula	Old shield with lower average deformation	Koyna reservoir-associated; Latur intraplate vulnerability
Indo-Gangetic plain	Foreland basin receiving Himalayan shaking	Soft alluvium, dense exposure, amplification/liquefaction

[FACT] The 2001 Bhuj earthquake demonstrates severe intraplate risk in Kachchh and the role of non-engineered construction and liquefaction.

[FACT] The 2011 Sikkim earthquake illustrates Himalayan/NE mountain cascades, including landslides and access disruption. The 2015 Gorkha earthquake in Nepal affected parts of northern and eastern India, showing transboundary Himalayan risk.

[FACT] The Peninsular shield is relatively stable, not aseismic. Latur and Koyna are essential close-option corrections.

[CA FACT - PIB/NCS, 22 July 2026] NCS reported 16 earthquakes in the Kangra-Chamba region during June 2026, including one in the M5.0-5.9 range, and stated that its National Seismological Network had 172 stations. [INFERENCE] Event counts show monitoring and active deformation, not a deterministic forecast of a larger event.

> **ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM:** India's earthquake geography must connect tectonic source, geomorphic site and settlement vulnerability; a zone label alone cannot explain loss.

CLOSING RECALL FLOW - INDIA'S TECTONIC FRAMEWORK AND REGIONAL SEISMIC RISK

```
START / CONCEPT: INDIA'S TECTONIC FRAMEWORK AND REGIONAL SEISMIC RISK
        |
        v
EXACT TERMS: India'S Tectonic Framework · Regional Seismic Risk · Himalaya · Ongoing continent-
continent shortening and thrusting · Steep slopes, road isolation, vulnerable hill construction ·
NE/Indo-Burmese
        |
        v
MECHANISM / ARGUMENT: The 2001 Bhuj earthquake demonstrates severe intraplate risk in Kachchh and
the role of non-engineered construction and liquefaction.
        |
        v
CONSEQUENCE / CONTRAST: The Peninsular shield is relatively stable, not aseismic.
        |
        v
UPSC TRAP / ANSWER-USE: Do not miss this limiting distinction: latur and Koyna are essential close-
option corrections.
        |
        v
ANSWER-GRABBING FORMULATION: India's earthquake geography must connect tectonic source, geomorphic
site and settlement vulnerability; a zone label alone cannot explain loss.
```

SESSION 18 - INDIA SEISMIC ZONATION, MICROZONATION AND CLASSIFICATION CAUTION

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: NCS defines seismic hazard microzonation as classification into zones of relatively similar exposure to earthquake effects and describes it as a GIS-based, site-specific planning tool.

Technical definition: Microzonation must feed building regulation, land use, lifeline siting and retrofit priorities.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> NCS defines seismic hazard microzonation as classification into zones of relatively similar exposure to earthquake effects and describes it as a GIS-based, site-specific planning tool.

MUST-WRITE KEYWORDS

- **India Seismic Zonation**
- **Microzonation**
- **Classification Caution**
- **not**
- **Zone II**
- **Lower national-scale design hazard**

How to use them: Frame the answer through India Seismic Zonation; define Microzonation, connect Classification Caution with not to explain the mechanism, and use Zone II for the decisive comparison or qualification.

What the familiar four-zone framework means

Publicly verifiable BIS design-code baseline	Broad description
Zone II	Lower national-scale design hazard
Zone III	Moderate
Zone IV	High
Zone V	Very high

[FACT] The familiar BIS macrozonation uses Zones II, III, IV and V; there is no Zone I in that framework. Zone V includes very-high-hazard sectors such as parts of the Himalaya/NE, Kachchh and Andaman-Nicobar.

[FACT] NCS defines seismic hazard microzonation as classification into zones of relatively similar exposure to earthquake effects and describes it as a GIS-based, site-specific planning tool. Its public page records Delhi microzonation at 1:10,000 scale and work in other cities.

Official-classification caution at the 23 August 2026 cutoff

Official strand	What it says	Exam-safe treatment
Publicly searchable BIS design baseline	IS 1893 (Part 1):2016 remains the publicly verifiable standard; familiar map is Zones II-V	Use for code-baseline answers; verify latest edition/amendment before engineering application
PIB/MoES parliamentary reply, 17 Dec 2025	Refers to an updated zonation map placing the Himalayan arc in a new Zone VI	Cite with source/date as an official hazard-map revision claim
PIB/MoES parliamentary reply, 22 Jul 2026	Describes Kangra-Chamba-Dharamshala as Zone V, "the highest earthquake hazard zone in India"	Shows official public communication is not yet internally harmonised

[LIMIT] Because the corresponding adopted BIS standard/map could not be independently verified on the public BIS portal at the package cutoff, this package does **not** silently replace the code baseline with Zone VI. It teaches both official strands and tells the learner to state the source and date.

```
MACROZONATION                        MICROZONATION
national/regional scale              city/local scale
        |                                    |
broad design hazard                  soil + basin + fault + groundwater
        |                                    |
cannot locate next event             cannot predict next event either
        \_____________ both inform safer planning _____________/
```

[INFERENCE] A city-wide zone label can conceal high-risk reclaimed ground, old channels and soft basins next to safer rock sites. Microzonation must feed building regulation, land use, lifeline siting and retrofit priorities.

> **ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM:** Macrozonation tells a city the regional design problem; microzonation tells neighbourhoods how local ground modifies that problem.

CLOSING RECALL FLOW - INDIA SEISMIC ZONATION, MICROZONATION AND CLASSIFICATION CAUTION

```
START / CONCEPT: INDIA SEISMIC ZONATION, MICROZONATION AND CLASSIFICATION CAUTION
        |
        v
EXACT TERMS: India Seismic Zonation · Microzonation · Classification Caution · not · Zone II ·
Lower national-scale design hazard
        |
        v
MECHANISM / ARGUMENT: Because the corresponding adopted BIS standard/map could not be independently
verified on the public BIS portal at the package cutoff, this package does not silently replace the
code baseline with Zone VI.
        |
        v
CONSEQUENCE / CONTRAST: Microzonation must feed building regulation, land use, lifeline siting and
retrofit priorities.
        |
        v
UPSC TRAP / ANSWER-USE: Do not miss this limiting distinction: the familiar BIS macrozonation uses
Zones II, III, IV and V.
        |
        v
ANSWER-GRABBING FORMULATION: NCS defines seismic hazard microzonation as classification into zones
of relatively similar exposure to earthquake effects and describes it as a GIS-based, site-specific
planning tool.
```

SESSION 19 - INSTITUTIONS, RESILIENT CONSTRUCTION, EARLY WARNING LIMITS AND PREPAREDNESS

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Institutions, Resilient Construction, Early Warning Limits And Preparedness comprises Institutions, Resilient Construction and Early Warning Limits as its core connected dimensions.

Technical definition: Technically, Institutions, Resilient Construction, Early Warning Limits And Preparedness is analysed by relating Institutions to Resilient Construction, then testing the relationship through Early Warning Limits and Preparedness.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Institutions, Resilient Construction, Early Warning Limits And Preparedness comprises Institutions, Resilient Construction and Early Warning Limits as its core connected dimensions.

MUST-WRITE KEYWORDS

- **Institutions**
- **Resilient Construction**
- **Early Warning Limits**
- **Preparedness**
- **after rupture begins**
- **[CA FACT] PIB 22 July 2026**

How to use them: Frame the answer through Institutions; define Resilient Construction, connect Early Warning Limits with Preparedness to explain the mechanism, and use after rupture begins for the decisive comparison or qualification.

Institution	Core role in this topic
NCS, MoES	National seismic monitoring, event information, hazard and microzonation work
GSI	Geological/seismotectonic mapping and geoscientific evidence
BIS	Structural standards such as IS 1893 and allied earthquake-resistant design/construction/evaluation codes
NDMA/NIDM	National guidance, preparedness, training and capacity building
States/UTs/ULBs	Building approvals, bye-law enforcement, inspections, local land use and drills
INCOIS/ITEWC	Indian Ocean tsunami warning service

Resilient-building chain

```
SAFE SITE
  -> regular configuration + continuous load path
  -> ductile detailing + strong connections
  -> light/anchored non-structural elements
  -> quality materials and workmanship
  -> inspection and maintenance
  -> RETROFIT vulnerable existing stock and lifelines

Avoid: weak/soft storey, heavy unanchored parapets/tanks, plan irregularity,
poor masonry connections, open ground-floor discontinuity and code-without-enforcement.
```

[FACT - PIB, 23 July 2026] MoHUA stated that urban building approval and bye-law enforcement belong to ULBs/Urban Development Authorities; it cited Model Building Bye-Laws Chapter 6, BIS earthquake-resistant standards, and NDMA guidance on safer construction, structural strengthening and retrofitting, especially lifeline and critical infrastructure.

[FACT] Retrofitting may add confinement, bracing, shear walls, jacketing, improved connections or foundation measures depending on an engineering assessment. There is no universal retrofit recipe.

[FACT] Earthquake early warning detects an earthquake **after rupture begins** and may warn locations before the strongest waves arrive. It is not prediction; warning time is short and near-source blind zones are unavoidable.

[FACT] Practical community actions include Drop-Cover-Hold On during shaking, avoiding lifts, securing heavy objects, identifying safe assembly/evacuation routes, practising drills and checking utilities/fire hazards after shaking.

[INFERENCE] The implementation gap is often the decisive risk multiplier: a technically sound code that is not adopted, supervised or enforced does not create a safe building.

Current-affairs anchor box - static/current kept separate

- **[CA FACT] NCS June 2026 monthly report:** 445 earthquakes were located globally by NCS during the month, with 230 in India and its neighbourhood; a preliminary report covered the 5 June Chamba M5.0 event.
- **[CA FACT] PIB 22 July 2026:** 16 June events in Kangra-Chamba; NCS network stated as 172 stations.
- **[CA FACT] PIB 23 July 2026:** structural-safety audits, BIS codes, retrofitting and state/ULB enforcement responsibilities.
- **[CA FACT] USGS reviewed events:** Mw 7.8 Philippines on 7 June 2026 and Mw 7.7 north-northwest of Ende, Indonesia, on 14 August 2026 illustrate continuing Ring-of-Fire seismicity.
- **[CA FACT] USGS/HVO 10 March 2026:** Kīlauea episode 43 provides a current volcanic-products and monitoring illustration.
- **[LIMIT]** None of these events predicts the timing of an Indian earthquake or proves a new BIS regulatory zone.

> **ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM:** Earthquake resilience is less a problem of knowing that a city is at risk than of converting that knowledge into enforced design, retrofitted lifelines and rehearsed behaviour.

CLOSING RECALL FLOW - INSTITUTIONS, RESILIENT CONSTRUCTION, EARLY WARNING LIMITS AND PREPAREDNESS

```
START / CONCEPT: INSTITUTIONS, RESILIENT CONSTRUCTION, EARLY WARNING LIMITS AND PREPAREDNESS
        |
        v
EXACT TERMS: Institutions · Resilient Construction · Early Warning Limits · Preparedness · after
rupture begins · [CA FACT] PIB 22 July 2026
        |
        v
MECHANISM / ARGUMENT: It is not prediction; warning time is short and near-source blind zones are
unavoidable.
        |
        v
CONSEQUENCE / CONTRAST: The implementation gap is often the decisive risk multiplier: a technically
sound code that is not adopted, supervised or enforced does not create a safe building.
        |
        v
UPSC TRAP / ANSWER-USE: Do not miss this limiting distinction: retrofitting may add confinement,
bracing, shear walls, jacketing, improved connections or foundation measures depending...
        |
        v
ANSWER-GRABBING FORMULATION: Institutions, Resilient Construction, Early Warning Limits And
Preparedness comprises Institutions, Resilient Construction and Early Warning Limits as its core
connected dimensions.
```

BASIC MCQS / REMEDIATION

Practice rule

These are original UPSC-style questions, not claimed PYQs. The renderer applies the stable `geography-03` seed to reposition the correct option while preserving its content. Final keys must be balanced and non-patterned.

MCQ 1 - Endogenic processes

Which statement most accurately describes an endogenic process?

A. It is driven mainly by energy and forces originating within the Earth and includes tectonism, earthquakes and vulcanism. B. It is any process that lowers relief, including only rivers, glaciers and wind. C. It refers exclusively to volcanic eruptions at plate boundaries. D. It is a climatic process caused by unequal solar heating.

Answer: A.

Explanation: Endogenic processes are internally driven. Option B describes exogenic denudation, C is too narrow because earthquakes and non-volcanic tectonism also qualify, and D belongs to atmospheric energy processes.

MCQ 2 - Plate-boundary signature

Which combination is correctly matched?

A. Subduction margin - trench, inclined shallow-to-deep seismic zone and volcanic arc B. Transform margin - deep-focus earthquakes and continuous arc volcanism C. Mid-ocean ridge - predominantly deep earthquakes and rhyolitic calderas D. Continent-continent collision - ocean trench and routine basaltic island arc

Answer: A.

Explanation: A gives the diagnostic subduction system. Transform and ridge earthquakes are mainly shallow. A mature continent-continent collision has consumed the intervening oceanic slab and is not a standard island-arc setting.

MCQ 3 - Normal fault

A normal fault most directly indicates:

A. Crustal compression in which the hanging wall moves up B. Horizontal shear with no vertical component by definition C. Magma-chamber collapse without brittle failure D. Crustal extension in which the hanging wall moves down relative to the footwall

Answer: D.

Explanation: Tensional stress lengthens crust and produces normal faulting. A defines reverse faulting, B describes ideal strike-slip movement, and C describes a caldera process rather than a fault definition.

MCQ 4 - Elastic rebound

Elastic rebound theory explains:

A. Why all aftershocks are smaller than every foreshock B. How slow strain accumulation on a locked fault can be released by sudden rupture and partial recovery C. Why liquid magma necessarily triggers a tectonic earthquake D. How surface waves convert into tides in the ocean

Answer: B.

Explanation: The theory links gradual loading to sudden slip. A is false because sequence labels are retrospective and magnitudes vary. C and D conflate unrelated processes.

MCQ 5 - Earthquake source class

Which event is best classified as reservoir-associated seismicity?

A. Slip promoted near susceptible faults after reservoir loading and pore-pressure change B. A megathrust rupture caused solely by atmospheric pressure C. A cave-roof collapse generating a basin-wide tsunami D. A volcanic ash cloud recorded by a seismometer

Answer: A.

Explanation: Reservoir impoundment can alter loading and pore pressure near pre-existing weaknesses. It does not create plate stress from nothing. The other options confuse source mechanisms and scales.

MCQ 6 - Focal depth

Which statement about focal depth is correct?

A. A shallow earthquake is always more damaging than any deep earthquake. B. Focal depth and epicentral distance are the same quantity. C. Transform faults routinely generate earthquakes at 500-700 km depth. D. Deep-focus earthquakes are characteristic of descending slabs and may occur hundreds of kilometres below the surface.

Answer: D.

Explanation: Intermediate and deep seismicity follows subducting lithosphere. Transform events are shallow. Damage cannot be inferred from depth alone, and depth is vertical source position rather than surface distance.

MCQ 7 - Focus and epicentre

Which distinction is correct?

A. The focus is the worst-damaged town; the epicentre is the largest-magnitude station. B. The focus is the subsurface rupture origin; the epicentre is its vertical projection on the surface. C. The epicentre lies inside the Earth and the focus lies on the surface. D. Both terms refer to the entire seismic zone.

Answer: B.

Explanation: A is the standard geometric distinction. Damage may peak away from the epicentre because of directivity, site effects and construction.

MCQ 8 - Locating an epicentre

Why are observations from at least three seismic stations used in the elementary epicentre-location method?

A. Each station supplies a distance circle, and their intersection estimates the epicentre. B. One station measures longitude, one latitude and one magnitude. C. S waves alone indicate a unique direction at each station. D. Three stations eliminate the need for an Earth-velocity model.

Answer: A.

Explanation: P-S arrival difference gives distance, not a unique bearing. Multiple circles constrain position. Operational location uses more stations and a velocity model.

MCQ 9 - P and S waves

Which statement is correct?

A. S waves arrive before P waves because shear motion is faster. B. P waves move particles only perpendicular to propagation. C. Both P and S waves are confined to the Earth's surface. D. P waves are compressional and can cross liquids, whereas S waves are shear waves requiring a solid medium.

Answer: D.

Explanation: P waves arrive first and travel through solids and liquids. S waves are slower and cannot propagate through liquid. Both are body waves.

MCQ 10 - Surface waves

Which pairing is correct?

A. Love wave - liquid-only; Rayleigh wave - gas-only B. Love wave - mainly horizontal shear; Rayleigh wave - elliptical rolling motion C. Love wave - first arrival; Rayleigh wave - always second arrival D. Love wave - compressional body wave; Rayleigh wave - deep-core shear wave

Answer: B.

Explanation: Love and Rayleigh are surface-wave modes. Love produces strong horizontal motion; Rayleigh produces rolling vertical and horizontal motion.

MCQ 11 - Shadow zones

The absence of direct S waves beyond about 103° from an earthquake primarily indicates that:

A. The outer core is liquid and lacks the shear rigidity needed for S-wave transmission. B. The mantle is entirely molten. C. The inner core is a vacuum. D. S waves travel faster than P waves through liquids.

Answer: A.

Explanation: S-wave extinction is the decisive liquid outer-core evidence. It does not imply a molten mantle or empty inner core.

MCQ 12 - Magnitude and intensity

Which statement is correct?

A. Intensity is constant everywhere for a given event. B. One earthquake has one preferred magnitude estimate but may produce many place-specific intensities. C. Magnitude is assigned from building damage alone. D. Mw and MMI are interchangeable labels for the same quantity.

Answer: B.

Explanation: Magnitude describes source size; intensity describes effects at a location. Mw is a magnitude scale, while MMI is an intensity scale.

MCQ 13 - Moment magnitude

Moment magnitude is physically linked most directly to:

A. Number of media reports and casualties B. Rock rigidity, ruptured fault area and average slip C. Distance from the nearest city only D. Duration of the monsoon

Answer: B.

Explanation: Seismic moment is rigidity × area × slip. Human impact does not define Mw.

MCQ 14 - Global belts

Which global belt is correctly characterised?

A. Circum-Pacific belt - subduction trenches, volcanic arcs, shallow-to-deep earthquakes and tsunami potential B. Mid-Atlantic Ridge - continent-continent collision with no volcanism C. Mediterranean-Himalayan belt - a single uniform oceanic island arc D. Stable shields - completely earthquake-free interiors

Answer: A.

Explanation: A captures the Ring of Fire system. Ridges are divergent and volcanic; the Mediterranean-Himalayan belt is tectonically complex; shields can host intraplate earthquakes.

MCQ 15 - Site amplification

Site amplification most commonly occurs when:

A. A volcano emits only water vapour. B. A fault is absent from the entire region. C. Seismic waves enter softer unconsolidated sediment and selected motions increase relative to nearby rock sites. D. All waves accelerate through soft soil and lose amplitude completely.

Answer: C.

Explanation: Soft sediment and basin geometry can amplify and prolong shaking. The response is frequency-dependent, not a universal simple increase.

MCQ 16 - Liquefaction

Liquefaction is best described as:

A. Dissolution of limestone by rainwater B. Conversion of lava into volcanic glass C. Melting of bedrock due to frictional heat D. Temporary strength loss in loose saturated granular sediment as pore pressure rises during shaking

Answer: D.

Explanation: Liquefaction reduces effective stress; the grains do not literally melt. Floodplains, deltas, reclaimed land and old channels are common susceptible settings.

MCQ 17 - Hazard and disaster

Which formulation is most accurate?

A. Every recorded earthquake is automatically a disaster. B. Vulnerability is the magnitude of the earthquake source. C. A seismic hazard becomes a disaster when it causes severe disruption in an exposed, vulnerable society beyond coping capacity. D. Capacity increases risk by definition.

Answer: C.

Explanation: Hazard, exposure, vulnerability and capacity must be separated. Small uninhabited events may be hazards without disasters.

MCQ 18 - Tsunami source

Which earthquake mechanism is generally most efficient at generating a large basin-wide tsunami?

A. A deep continental strike-slip event with no water displacement B. A minor mine collapse far inland C. A slow atmospheric-pressure change D. A submarine megathrust producing rapid vertical displacement of the sea floor

Answer: D.

Explanation: Rapid vertical displacement of a large sea-floor area moves the whole water column. Strike-slip movement is less efficient unless it triggers a landslide or includes vertical displacement.

MCQ 19 - Tsunami behaviour

As a tsunami enters shallower coastal water, it generally:

A. Converts into an S wave B. Speeds up, lengthens and necessarily disappears C. Slows, shortens in wavelength and grows in height through shoaling D. Becomes a tide controlled by the Moon

Answer: C.

Explanation: Reduced depth lowers speed; energy compression raises wave height. A tsunami is not a tidal wave.

MCQ 20 - Volcano anatomy

Which distinction is correct?

A. A crater is a vent-centred depression; a caldera is a much larger collapse depression commonly linked to substantial magma withdrawal. B. A caldera is always a small side vent. C. A magma chamber is located only above the crater. D. A conduit is an erosional river channel.

Answer: A.

Explanation: Caldera collapse is structurally and dimensionally distinct from an ordinary crater. Chambers and conduits are subsurface parts of the magma system.

MCQ 21 - Silica and viscosity

All else equal, increasing magma silica commonly:

A. Guarantees a harmless effusive eruption B. Eliminates dissolved volatiles C. Converts magma into sedimentary rock before ascent D. Increases polymerisation and viscosity, making gas escape more difficult

Answer: D.

Explanation: Silica-rich structures impede flow and degassing. Composition controls tendency, not certainty.

MCQ 22 - Explosive eruption

Which chain most directly favours explosive fragmentation?

A. No volatiles -> increasing gas pressure B. Viscous magma -> inhibited degassing -> gas expansion during ascent -> fragmentation C. Falling pressure -> gases remain permanently dissolved D. Fluid basalt -> open degassing -> guaranteed Plinian column

Answer: B.

Explanation: Exsolution and trapped gas create overpressure. Fluid magma with open pathways usually degasses more easily.

MCQ 23 - Dyke and sill

Which pairing is correct?

A. Dyke - surface ash deposit; sill - tsunami wave B. Dyke - concordant dome; sill - deep batholith only C. Dyke - cuts across layers; sill - intrudes approximately parallel to layers D. Dyke - crater; sill - caldera

Answer: C.

Explanation: Relation to country-rock layering is the close-option distinction.

MCQ 24 - Volcanic products

Which one is correctly matched?

A. Pyroclastic density current - cool groundwater moving slowly uphill B. Lahar - water-mobilised volcanic debris flow, commonly channelled down valleys C. Lapilli - volcanic gas only D. Volcanic bomb - ash particle smaller than 2 mm

Answer: B.

Explanation: Lahars are sediment-water flows. Density currents are hot gas-particle flows. Lapilli are 2-64 mm fragments; bombs/blocks are coarse.

MCQ 25 - Eruption style

Which eruption style is most closely associated with sustained high eruption columns and widespread tephra?

A. Icelandic fissure B. Quiet effusive degassing only C. Plinian D. Hawaiian

Answer: C.

Explanation: Plinian activity is highly explosive. Hawaiian and Icelandic styles are generally more effusive.

MCQ 26 - Volcano morphology

Which description best fits a shield volcano?

A. A steep cone made only of glacial ice B. Broad gentle slopes built mainly by repeated low-viscosity basaltic flows C. A fold mountain produced without magma D. A deep ocean trench

Answer: B.

Explanation: Fluid lava travels far and builds a low-angle profile. Composite volcanoes are generally steeper and layered.

MCQ 27 - Activity status

Which statement is correct?

A. Active, dormant and extinct are evidence-based status descriptions, not a compulsory one-way life cycle. B. A shield volcano can never be active. C. Dormant means eruption is physically impossible. D. Every volcano inactive for one century is automatically extinct.

Answer: A.

Explanation: Geological, geophysical and gas evidence matters. Long repose may still end in renewed activity.

MCQ 28 - Ring of Fire

The Circum-Pacific “Ring of Fire” is best understood as:

A. A perfect circle of only basaltic shield volcanoes B. A climatic belt created by ocean currents C. A continent-interior rift with no earthquakes D. A linked system of trenches, subducting slabs, earthquake zones and volcanic arcs around much of the Pacific margin

Answer: D.

Explanation: The Ring of Fire is a geophysical system, not a mathematically perfect ring or a single volcano type.

MCQ 29 - Divergent volcanism

Magma at a mid-ocean ridge is generated mainly because:

A. Continents collide and completely stop mantle flow B. Surface ash is buried by glaciers C. A cold slab adds water to a mantle wedge D. Hot mantle rises and partially melts as pressure decreases

Answer: D.

Explanation: Decompression melting is the principal ridge mechanism. Slab-derived flux melting describes subduction arcs.

MCQ 30 - Subduction volcanism

At a subduction zone, magma generation is promoted when:

A. The oceanic plate becomes less dense than the atmosphere B. Volatiles released from the descending slab lower the melting temperature in the overlying mantle wedge C. The trench removes all water from minerals D. The transform fault creates a mid-ocean ridge

Answer: B.

Explanation: Flux melting above the slab feeds volcanic arcs.

MCQ 31 - Mantle plume

Which is the characteristic plume/hot-spot prediction?

A. A plume tail always creates continent-continent collision. B. Hot-spot chains become younger away from the active end in both directions. C. Every hot spot lies exactly on a trench. D. A moving plate over a relatively persistent source can create an age-progressive volcanic chain.

Answer: D.

Explanation: Age generally increases away from the active hot-spot end. The hypothesis is useful but scientifically qualified.

MCQ 32 - Flood basalt

Which statement about the Deccan Traps is correct?

A. They consist only of limestone caves. B. They are a continental flood-basalt province built mainly by extensive fissure eruptions, not a single giant cone. C. They were formed by a modern tsunami. D. They are an active Himalayan island arc.

Answer: B.

Explanation: Repeated basalt sheets produce plateau and step-like relief.

MCQ 33 - Volcanic climate forcing

Temporary global or hemispheric cooling after a major explosive eruption is most directly associated with:

A. Stratospheric sulphate aerosols reflecting incoming sunlight B. Coarse bombs remaining suspended for decades C. Lava absorbing all ocean heat D. S waves entering the atmosphere

Answer: A.

Explanation: Sulphur gases converted to fine aerosols in the stratosphere provide the main short-term radiative pathway.

MCQ 34 - Constructive effect

Which is a constructive long-term effect of vulcanism?

A. Guaranteed absence of future hazards B. Elimination of all atmospheric gases C. Conversion of every volcanic slope into a desert D. Creation of new crust and land, geothermal potential, hydrothermal minerals and weathered fertile material

Answer: D.

Explanation: Benefits coexist with episodic risk; they do not remove it.

MCQ 35 - India tectonic setting

Which regional pairing is correct?

A. Kachchh - mid-ocean ridge spreading axis B. Himalaya - routine oceanic island arc after complete subduction C. Andaman-Nicobar - Sunda subduction/island-arc setting with earthquake, tsunami and volcanic links D. Peninsular India - absolutely aseismic shield

Answer: C.

Explanation: Andaman is the subduction/island-arc system. Kachchh is an intraplate rift-reactivation setting, Himalaya a continental collision, and the peninsula is relatively stable but not aseismic.

MCQ 36 - Bhuj lesson

The 2001 Bhuj earthquake is most useful for demonstrating:

A. That all major Indian earthquakes occur only in the Himalaya B. That liquefaction occurs only in solid granite C. Severe intraplate risk in Kachchh combined with vulnerable buildings and local ground effects D. That seismic zones predict event dates

Answer: C.

Explanation: Bhuj corrects the boundary-only and stable-shield assumptions and illustrates the risk equation.

MCQ 37 - Peninsular exceptions

Which pair corrects the statement “Peninsular India is earthquake-free”?

A. Kīlauea and Mauna Loa B. Andes and Aleutians C. Barren Island and Narcondam D. Koyna and Latur

Answer: D.

Explanation: Koyna and Latur are standard Indian intraplate/reservoir-associated examples.

MCQ 38 - Seismic zonation

Which statement about seismic zonation is correct?

A. It guarantees equal shaking everywhere inside one zone. B. It replaces structural design and construction quality. C. It is a broad hazard/design classification and does not predict the time, place and magnitude of the next earthquake. D. It is identical to an intensity map of one past event.

Answer: C.

Explanation: Macrozonation supports design but cannot resolve local site response or predict events.

MCQ 39 - Microzonation

Seismic microzonation primarily adds:

A. Finer site-specific information on geology, soil, ground motion, liquefaction and related local effects B. A list of volcanic eruption styles C. A guaranteed earthquake date for each ward D. A replacement for all building codes

Answer: A.

Explanation: It is a local hazard/risk-planning tool, not prediction.

MCQ 40 - Early warning

Earthquake early warning differs from prediction because it:

A. names the exact future event years before rupture. B. prevents the earthquake from occurring. C. Detects a rupture that has already begun and may alert more distant places before strong shaking arrives. D. works equally well at the epicentre with unlimited warning time.

Answer: C.

Explanation: Warning time is short and near-source blind zones remain. EEW can automate protective action but cannot stop the source.

MCQ 41 - Remedial: magnitude versus intensity

A learner writes, “The earthquake had intensity Mw 7.0 everywhere.” What is the best correction?

A. Intensity and magnitude are identical, but the number must be rounded. B. Mw is a magnitude; intensity is place-specific and should be expressed through an intensity scale or observed effects. C. Mw measures only casualties. D. Every place has the same intensity for one event.

Answer: B.

Explanation: The sentence mixes event size and local effect. One event may produce a mapped gradient of intensities.

MCQ 42 - Remedial: S waves

Why do S waves not pass through the liquid outer core?

A. Liquids always absorb P waves but transmit S waves. B. S waves are electromagnetic. C. S waves are tides. D. A liquid cannot sustain the shear stress required for S-wave propagation.

Answer: D.

Explanation: Shear rigidity is the controlling property.

MCQ 43 - Remedial: epicentre

Which statement avoids the epicentre trap?

A. The epicentre is always the place with the most deaths. B. The epicentre is a geometric surface projection; the greatest loss may occur elsewhere because site and vulnerability modify shaking. C. The epicentre is the entire fault plane. D. The epicentre measures magnitude.

Answer: B.

Explanation: Damage geography and source geometry must be separated.

MCQ 44 - Remedial: hazard and disaster

Which situation is a seismic hazard without necessarily being a disaster?

A. Any earthquake listed by a seismic network B. A building-code document C. A strong remote earthquake that causes no severe social disruption because exposure is minimal D. A dormant fault with no possibility of movement

Answer: C.

Explanation: Disaster requires realised severe impact on society.

MCQ 45 - Remedial: tsunami terminology

Why is “tidal wave” an unsafe synonym for tsunami?

A. Tsunamis are generated by sudden water displacement and are not caused by astronomical tides. B. Tides are seismic body waves. C. Tsunamis cannot cross an ocean. D. Tsunamis occur only at low tide.

Answer: A.

Explanation: The source mechanism is unrelated to lunar/solar tides.

MCQ 46 - Remedial: shield versus composite

Which comparison is correct?

A. Shield volcanoes are always extinct and composite volcanoes always active. B. Both terms classify focal depth. C. Shield volcanoes are broad products of fluid flows; composite volcanoes are layered cones commonly associated with mixed and explosive activity. D. Composite volcanoes form only at mid-ocean ridges.

Answer: C.

Explanation: Form and status are separate classifications.

MCQ 47 - Remedial: zone map

A city lies in one national seismic zone. Which conclusion is justified?

A. Every neighbourhood has identical soil and shaking. B. The next earthquake date is known. C. The zone supplies a broad design-hazard input, but local microzonation and code compliance are still required. D. Retrofitting is unnecessary.

Answer: C.

Explanation: Macrozones do not erase local geology or building differences.

MCQ 48 - Remedial: prediction versus preparedness

Which policy follows from the present inability to predict earthquake time deterministically?

A. Prioritise enforced resilient construction, retrofit, lifeline continuity, monitoring, drills and public education. B. Build only after an earthquake warning is issued. C. Treat low recent seismicity as proof of permanent safety. D. Stop seismic monitoring.

Answer: A.

Explanation: Risk reduction manages consequences under uncertainty; absence of recent events does not erase accumulated hazard.

PYQS AND ANSWER PRACTICE

PYQ control note

- Exact wording is reproduced from the routed local UPSC question papers.
- UPSC does not publish model Mains answers; the Mains solutions below are independent examiner-grade models.
- 2018 and 2023 Prelims keys are unavailable in the local verified key set; answers are explicitly labelled inferred.
- The 2024 Set-A answer is verified from the local official key.
- The 2026 Set-A key is local and provisional. It is reported as provisional, never as the final UPSC key and never replaced by an inference.

VERIFIED PYQ 1 - UPSC Prelims 2018 GS-I Q85: Barren Island

Exact question

Consider the following statements:

1. The Barren Island volcano is an active volcano located in the Indian territory.
2. Barren Island lies about 140 km east of Great Nicobar.
3. The last time the Barren Island volcano erupted was in 1991 and it has remained inactive since then.

Which of the statements given above is/are correct?

A. 1 only B. 2 and 3 only C. 3 only D. 1 and 3 only

INFERRED ANSWER - NOT OFFICIALLY VERIFIED LOCALLY: A, 1 only. Confidence: high.

Explanation

- Statement 1 is correct: Barren Island in Indian territory is the country's active-volcano example.
- Statement 2 is false: the location relative to Great Nicobar is wrong; Barren Island lies in the northern Andaman Sea, northeast of Port Blair, not about 140 km east of Great Nicobar.
- Statement 3 is false: activity resumed after 1991; therefore “remained inactive since then” is incorrect.

Why this PYQ matters: UPSC combines activity status, map location and eruption chronology. “Active” is not equivalent to “erupting continuously.”

VERIFIED PYQ 2 - UPSC Prelims 2023 GS-I Q65: P and S waves

Exact question

Consider the following statements:

1. In a seismograph, P waves are recorded earlier than S waves.
2. In P waves, the individual particles vibrate to and fro in the direction of wave propagation, whereas in S waves, the particles vibrate up and down at right angles to the direction of wave propagation.

Which of the statements given above is/are correct?

A. 1 only B. 2 only C. Both 1 and 2 D. Neither 1 nor 2

INFERRED ANSWER - NOT OFFICIALLY VERIFIED LOCALLY: C, both 1 and 2. Confidence: high.

Explanation

P waves are faster compressional body waves, so they arrive first. Their particle motion is parallel to propagation. S waves are shear waves with particle motion perpendicular to propagation. The statement's “up and down” language illustrates the transverse relation; S polarisation may occur in different perpendicular orientations.

VERIFIED PYQ 3 - UPSC Prelims 2024 GS-I Q3: products of eruptions

Exact question

Consider the following:

1. Pyroclastic debris
2. Ash and dust
3. Nitrogen compounds
4. Sulphur compounds

How many of the above are products of volcanic eruptions?

A. Only one B. Only two C. Only three D. All four

OFFICIALLY VERIFIED LOCAL SET-A KEY: D, all four.

Explanation

Explosive eruptions produce pyroclastic debris, ash and dust. Volcanic emissions include sulphur compounds and can include nitrogen species/compounds. The official key controls the objective answer. The elimination lesson is that eruption products include gases and chemical species, not only visible lava and rock fragments.

VERIFIED PYQ 4 - UPSC Prelims 2026 GS-I Q31: Tungurahua

Exact question

Tungurahua Volcano, which was declared a Global Geopark by UNESCO in 2025, is situated in which one among the following countries?

A. Ecuador B. Peru C. Bolivia D. Colombia

LOCAL PROVISIONAL SET-A KEY: A, Ecuador. Final official UPSC key not verified.

Status explanation: UNESCO's official 2025 material identifies Tungurahua Volcano UNESCO Global Geopark in Ecuador. The local answer-key document is expressly provisional; this package records its status and does not infer or upgrade it to a final key.

Map trap: Ecuador and Colombia are both Andean states on the Ring of Fire. Tungurahua belongs to Ecuador's volcanic arc.

VERIFIED PYQ 5 - UPSC Mains GS-I 2018 Q6, 10 marks

Exact question: “Define mantle plume and explain its role in plate tectonics.” *(Answer in 150 words.)*

Demand decoding

“Define” requires plume anatomy; “explain its role” requires hot-spot chains, flood basalts and the plate-motion reference frame. A generic answer on plate boundaries misses the demand.

Model answer

A **mantle plume** is a hypothesised buoyant upwelling of anomalously hot mantle material that rises substantially independently of ordinary plate-boundary circulation.

```
deep thermal anomaly -> broad plume head -> flood basalt
                     -> persistent tail -> hot spot
moving plate over tail -> age-progressive volcanic chain
```

Its role in plate tectonics is threefold.

1. **Intraplate volcanism:** It explains volcanoes away from plate margins, such as the Hawaiian hot-spot chain.

2. **Large igneous provinces:** Arrival of a broad plume head may generate voluminous fissure eruptions; the Deccan Traps are the Indian illustration.
3. **Plate-motion record:** Progressive ageing away from an active hot spot indicates the direction and approximate relative rate of plate movement, independently checking evidence from magnetic stripes.
4. **Rifting link:** Thermal doming and magmatism may assist continental break-up, although plate stresses also matter.

The hypothesis is not a complete replacement for plate tectonics. Plumes add a vertical mantle component to a system dominated by horizontal plate interaction, and some intraplate volcanism may have shallower causes.

Why this earns marks

It gives a direct definition, one process diagram, two named examples, four roles and a scientific qualification instead of describing ordinary arc volcanoes.

VERIFIED PYQ 6 - UPSC Mains GS-I 2020 Q4, 10 marks

Exact question: "Discuss the geophysical characteristics of Circum-Pacific Zone." *(Answer in 150 words.)*

Demand decoding

"Geophysical characteristics" requires tectonic structure, earthquake-depth pattern, volcanism, landforms and tsunami potential-not a list of Pacific countries.

Model answer

The Circum-Pacific Zone or **Ring of Fire** is the world's dominant linked belt of subduction-related seismicity and volcanism around much of the Pacific margin.

```
oceanic plate -> trench -> inclined Wadati-Benioff seismic zone
                          -> mantle-wedge melting -> volcanic arc
```

Its principal characteristics are:

- **Deep trenches** such as Peru-Chile, Japan and Mariana mark descending plates.
- **Shallow, intermediate and deep earthquakes** outline subducting slabs; powerful megathrust events occur on locked plate interfaces.
- **Volcanic arcs** form as slab-derived volatiles promote mantle-wedge melting. Continental arcs occur in the Andes/Cascades and island arcs in Japan, the Philippines and Aleutians.
- **Explosive stratovolcanism** is common because arc magmas are often volatile-rich and relatively viscous.
- **Tsunami potential** is high where submarine megathrust rupture displaces the sea floor.
- Transform segments and back-arc basins add local complexity.

Thus, the Ring is not merely a concentration of volcanoes but a trench-slab-earthquake-arc system produced by continuing Pacific-margin convergence.

Why this earns marks

The answer reconstructs the entire geophysical system, uses map-ready examples and ends with a mechanism-led conclusion.

VERIFIED PYQ 7 - UPSC Mains GS-I 2021 Q7, 10 marks

Exact question: "Mention the global occurrence of volcanic eruptions in 2021 and their impact on regional environment." *(Answer in 150 words.)*

Demand decoding

The word "mention" requires multiple named occurrences, but "impact on regional environment" requires a short mechanism and consequence for each rather than a bare list.

Model answer

Volcanic eruptions in 2021 occurred across several tectonic settings, especially subduction arcs and hot spots.

Occurrence	Setting/product	Regional environmental impact
La Soufrière, St Vincent	Explosive island-arc eruption	Ash affected air, water, crops and inter-island activity; pyroclastic and lahar hazards threatened valleys
Nyiragongo, DR Congo	Fluid lava from the East African Rift region	Lava affected settlements and infrastructure around Goma; evacuation and seismic unrest disrupted livelihoods
Cumbre Vieja, La Palma, Spain	Intraplate Canary hot-spot volcanism	Lava buried settlements and plantations; ash disrupted air quality and aviation
Semeru, Indonesia	Subduction-arc stratovolcano	Pyroclastic flows, ash and rain-remobilised debris affected valleys and communities

The impacts were therefore spatially selective: wind controlled ash, relief channelled density currents and lahars, and island/urban exposure magnified disruption. Yet, over longer periods, volcanic materials may create new land and fertile weathered soils. Regional impact depends as much on settlement, drainage and wind as on eruption magnitude.

Why this earns marks

It names globally distributed examples, ties each to a product/pathway and provides the constructive counterpoint without diluting the hazard analysis.

SUPPLEMENTARY CROSS-OWNED PYQ 8 - UPSC Mains GS-I 2025 Q7, 10 marks

Exact question: “What are Tsunamis? How and where are they formed? What are their consequences? Explain with examples.” *(Answer in 150 words.)*

Ownership note: The institutional disaster-management route is owned elsewhere, but the physical geography is directly relevant and therefore solved here.

Model answer

A **tsunami** is a train of long-period gravity waves generated by sudden displacement of a large water column; it is not a tide or ordinary wind wave.

Formation and location: Submarine megathrust earthquakes at subduction zones are the chief cause. Rapid vertical sea-floor movement displaces water, producing low-amplitude, fast waves in the deep ocean. As they enter shallow water, speed and wavelength fall while height increases through shoaling. Volcanic collapse, submarine landslides and rare impacts can also generate tsunamis. Major source belts include the Pacific Ring of Fire and the Sunda-Andaman subduction zone.

Consequences: Run-up and inundation cause drowning, erosion, debris impact and infrastructure loss; saline water contaminates soils and groundwater; ports, fisheries, tourism and coastal livelihoods suffer. Funnel-shaped bays, low slopes and dense unplanned settlements magnify loss.

The 2004 Sumatra-Andaman earthquake devastated Indian Ocean coasts; the 2011 Tōhoku event demonstrated earthquake-tsunami-nuclear cascading risk. Warning systems reduce exposure, but near-source coasts must also use natural warning signs and rapid evacuation.

Why this earns marks

Every sub-demand-definition, how, where, consequences and examples-is visibly answered.

ORIGINAL 10-MARK QUESTION 1

Question: Explain why a moderate earthquake may cause greater damage than a larger earthquake. *(150 words.)*

Model answer

Magnitude measures source size, not automatic damage. A moderate event may be more destructive when the complete risk chain is unfavourable:

1. **Shallow source and proximity:** less energy is lost before waves reach a settlement.
2. **Directivity:** rupture may focus strong motion towards the city.
3. **Soft ground:** alluvium and basin sediments amplify selected frequencies and prolong shaking.
4. **Liquefaction:** loose saturated sand loses strength, damaging foundations and utilities.
5. **Resonance:** dominant ground period may match vulnerable building heights.
6. **Weak building stock:** unreinforced masonry, soft storeys, poor connections and heavy unanchored elements fail.
7. **Exposure and cascading failure:** dense population, fire, blocked roads and lifeline disruption multiply loss.

```
damage ≠ magnitude alone
damage = source × path × site × exposure × vulnerability ÷ capacity
```

Thus a larger deep earthquake beneath rock and code-compliant construction may cause less loss than a smaller shallow event beneath saturated alluvium and non-engineered buildings. Earthquake policy must therefore combine source hazard with microzonation, resilient construction and retrofit.

ORIGINAL 10-MARK QUESTION 2

Question: Distinguish between volcanic ash, pyroclastic density currents and lahars, and explain why they require different hazard maps. *(150 words.)*

Model answer

All three contain volcanic material but move through different media and pathways.

Hazard	Composition/motion	Control	Hazard-map implication
Ash fall	Fine airborne particles settling from plume	Eruption-column height and wind	Broad downwind sectors; aviation, crops, water and roof load
Pyroclastic density current	Hot turbulent gas-particle mixture hugging ground	Collapse/blast energy and relief	Proximal flanks and valleys; extreme temperature and speed
Lahar	Water-mobilised ash and debris	Rain, snow/crater water and drainage	River valleys far beyond cone; can recur after eruption

Ash maps must model wind and accumulation thickness. Density-current maps require topographic run-out zones and exclusion areas. Lahar maps must follow drainage channels, bridges, settlements and reservoirs, including post-eruption rainfall scenarios.

Therefore, a single circular “volcano danger zone” is inadequate. Product, transport medium and terrain create distinct footprints; risk communication and evacuation must match each pathway.

ORIGINAL 15-MARK QUESTION 1

Question: Examine the tectonic and geomorphic reasons for the uneven distribution of earthquake risk in India. *(250 words.)*

Model answer

India's earthquake risk is uneven because **hazard sources vary by tectonic region**, while geomorphology and settlement alter the received shaking.

```
tectonic source -> wave path -> local ground -> exposed built environment
```

Tectonic differentiation

- **Himalaya:** continuing Indian-Eurasian convergence loads a broad fold-thrust belt. Shallow thrust earthquakes combine with steep slopes and landslide cascades.

- **North-East/Indo-Burmese region:** the eastern Himalayan syntaxis and Burma/Sunda interaction create complex compression and strike-slip deformation.
- **Andaman-Nicobar:** Sunda subduction creates earthquake, tsunami and volcanic multi-hazard.
- **Kachchh:** inherited rifts and faults can reactivate within the plate; Bhuj 2001 disproves the assumption that only boundaries generate severe risk.
- **Peninsular shield:** lower average deformation, but Koyna and Latur show reservoir-associated and intraplate exceptions.

Geomorphic and exposure differentiation

- Indo-Gangetic and urban-basin alluvium can amplify shaking.
- Loose saturated floodplain, delta and reclaimed deposits are liquefaction-prone.
- Himalayan relief triggers landslides and cuts roads, power and communications.
- Dense cities, vulnerable masonry, soft storeys and weak enforcement convert hazard into disaster.

Qualification: A BIS macrozone is a broad design input, not a prediction or a neighbourhood risk map. Microzonation, building inventory and lifeline analysis are necessary.

India's seismic-risk map is therefore not merely a tectonic map. It is a layered map of active structures, sediment, slope, urban exposure and construction quality.

ORIGINAL 15-MARK QUESTION 2

Question: “Volcanoes are both agents of destruction and foundations of regional development.” Discuss. *(250 words.)*

Model answer

Volcanoes produce acute, spatially differentiated hazards but also long-lived resources and land.

Destructive role

- Lava buries land and infrastructure.
- Pyroclastic density currents cause extreme proximal destruction.
- Ash damages crops, water, roofs, health and aviation over broad downwind areas.
- Lahars repeatedly affect valleys even after eruption.
- Volcanic gases create toxic local conditions; stratospheric sulphate aerosols may temporarily cool climate.
- Eruptions displace populations and disrupt island economies, ports and tourism.

Developmental foundations

- Weathered basalt and ash can form productive soils, as seen in the Deccan and Java, though fresh ash first imposes costs.
- Geothermal heat supports energy and heating in suitable regions.
- Hydrothermal systems concentrate metallic minerals and industrial materials.
- Volcanism constructs islands, plateaux and new crust.
- Distinctive landscapes support tourism and scientific education.

Analytical balance

The benefits are often continuous or recoverable, whereas hazards are episodic but potentially catastrophic. This explains continued settlement: communities exchange recurrent benefits for probabilistic risk. However, development becomes sustainable only when monitoring, exclusion zones, resilient infrastructure, evacuation routes and livelihood diversification prevent the resource advantage from becoming an exposure trap.

Thus volcanism is not “beneficial” or “destructive” in isolation. Regional outcome depends on product pathways, recurrence, governance and the capacity to live with a dynamic landscape.

ORIGINAL 20-MARK QUESTION 1

Question: Critically examine India's seismic-zonation framework as an instrument of urban risk reduction. *(250 words; expanded model supplied.)*

Model answer

Seismic zonation is indispensable because it converts national-scale seismotectonic evidence into a design language, but it is insufficient as a complete urban-risk instrument.

Value

1. **Common regulatory baseline:** BIS zonation and IS 1893 provide broad design-hazard parameters for structural standards.
2. **Prioritisation:** High-hazard regions can prioritise critical infrastructure, schools, hospitals and emergency routes.
3. **Mainstreaming:** A map makes earthquake resistance part of ordinary building regulation rather than post-disaster relief.
4. **Inter-regional consistency:** It enables comparable minimum design expectations across jurisdictions.

Limitations

1. **Scale:** A macrozone cannot resolve soft alluvium, reclaimed land, basin edges, active fault traces or liquefaction within a city.
2. **Hazard-risk gap:** It maps broad hazard, not exposure, building fragility, network dependence or social vulnerability.
3. **Boundary illusion:** Administrative users may treat zone lines as abrupt physical changes.
4. **Prediction misconception:** A zone neither predicts the next event nor guarantees uniform intensity.
5. **Implementation:** Weak local adoption, informal construction, poor supervision and ageing buildings can defeat a sound code.
6. **Current classification caution:** Official public communication in 2025-2026 refers inconsistently to Zone VI and Zone V for the Himalayan belt, while the publicly verifiable BIS design baseline remains the familiar II-V framework. Regulatory answers must cite the edition/date rather than assert an unverified settled change.

Way forward

```
macrozonation -> city microzonation -> parcel/geotechnical check
-> risk-sensitive land use -> code-compliant design
-> inventory + retrofit -> enforcement + audit + drills
```

NCS/MoES hazard work, GSI mapping, BIS codes, NDMA guidance and State/ULB enforcement must operate as one chain. Zonation is therefore the starting map of resilience, not its final guarantee.

ORIGINAL 20-MARK QUESTION 2

Question: Analyse the common tectonic logic and contrasting surface expressions of earthquakes, volcanoes and tsunamis around the Circum-Pacific belt. *(250 words; expanded model supplied.)*

Model answer

The Circum-Pacific belt is unified by plate convergence but differentiated by the way energy and material reach the surface.

Common tectonic logic

Oceanic lithosphere subducts beneath another oceanic plate or a continent. The locked interface accumulates elastic strain; rupture generates earthquakes. The descending slab releases volatiles, promoting mantle-wedge melting and volcanic arcs. Submarine megathrust displacement may move the ocean floor and generate tsunamis.

```
trench
  -> locked megathrust -> earthquake + possible vertical sea-floor displacement
  -> descending slab -> shallow/intermediate/deep seismicity
  -> slab volatiles -> mantle-wedge melt -> volcanic arc
```

Contrasting expressions

- **Earthquakes** are sudden elastic-energy release and occur along the interface, within the slab and in the overriding plate. Their map extends from shallow trench events to deep Wadati-Benioff zones.
- **Volcanoes** are mass and heat transfer. Magma rises landward of the trench, forming island arcs such as Japan/Philippines or continental arcs such as the Andes.
- **Tsunamis** are water-column responses. They require efficient vertical displacement or another rapid mass movement; not every large earthquake produces one.

Regional controls

Arc magma chemistry and gas favour explosive stratovolcanoes. Trench geometry, rupture area and depth influence earthquakes. Bathymetry, bays and coastal slope control tsunami run-up. Transform segments may be highly seismic but weakly volcanic.

Critical qualification

The “Ring” is neither continuous nor mechanically uniform. It is a connected plate-margin system whose hazards share a tectonic origin but require distinct monitoring, maps and preparedness.

OPTIONAL ADVANCED DEPTH - NOT REQUIRED FOR A CORE ANSWER

ADVANCED 1 - FINITE-FAULT RUPTURE, DIRECTIVITY AND FREQUENCY-DEPENDENT SITE RESPONSE

Optional enrichment: The point-source focus is useful for location, but a large earthquake ruptures a finite fault. Slip begins at the hypocentre and propagates. Where rupture travels towards a site, pulses can concentrate energy in the forward direction.

```
FAULT PLANE
hypocentre X ========> rupture propagation
                     city in forward direction may receive a strong velocity pulse
```

- **Directivity** helps explain elongated damage patterns that a simple epicentral circle cannot.
- **Frequency content** matters: short stiff structures respond differently from tall flexible structures.
- **Resonance** occurs when dominant ground periods approach a structure's natural period.
- **Basin-edge effects** can focus or convert waves; deep basins may prolong long-period shaking.
- **Nonlinearity:** very strong shaking can change soil stiffness and damping, so weak-motion amplification cannot always be extrapolated directly.

Core safeguard: These refinements improve a high-level answer but are not prerequisites for explaining magnitude, intensity or site amplification.

ADVANCED 2 - SEISMIC CYCLE, PALAEOSEISMOLOGY AND PROBABILITY

The “seismic cycle” is a conceptual sequence of interseismic loading, coseismic rupture and postseismic adjustment. It is not a clock.

Evidence	What it can add	Limitation
Historical catalogue	Past felt/damaging events	Short relative to geological recurrence
Instrumental record	Precise recent locations/magnitudes	Very short for rare great events
GPS/geodesy	Present deformation/strain accumulation	Does not give exact rupture date
Palaeoseismology	Ancient surface ruptures and recurrence ranges	Incomplete preservation and dating uncertainty
Probabilistic hazard assessment	Likelihood of ground motion in a time window	Probability is not prediction

Advanced answer line: A long quiet interval may indicate low activity, incomplete records or strain accumulation; it is not proof of safety.

ADVANCED 3 - CODE HIERARCHY, PERFORMANCE AND RETROFIT TRIAGE

Earthquake-resistant design does not mean "no damage." Modern performance thinking distinguishes:

```
frequent shaking -> limited/no significant damage
strong design shaking -> life safety, controlled damage
rare extreme shaking -> collapse prevention where feasible
```

Ductility allows controlled deformation and energy dissipation instead of brittle collapse. A continuous load path transfers inertial forces through floors, walls/frames and foundations.

Retrofit triage

1. Screen a building inventory.
2. Prioritise hospitals, schools, bridges, water, power and emergency facilities.
3. Conduct detailed structural/geotechnical evaluation.
4. Select structure-specific measures-jacketing, bracing, shear walls, diaphragm/connection repair, foundation improvement or non-structural anchorage.
5. Verify workmanship and maintain the intervention.

Limit: A generic checklist cannot replace a qualified structural assessment.

ADVANCED 4 - EARTHQUAKE EARLY WARNING AND OPERATIONAL LIMITS

An EEW system uses rapidly detected P waves and source estimates to issue alerts before stronger shaking reaches more distant users.

```
rupture begins -> nearest sensors detect P
-> automatic location/magnitude estimate
-> alert communication
-> trains slow | lifts stop | valves isolate | people protect

warning time = travel-time advantage - detection/processing/communication delay
```

Near the source, strong shaking may arrive before a useful alert. False/missed alerts, network density, telemetry, public understanding and automated end uses shape performance.

Advanced distinction: Forecasting changes probability over a period; prediction specifies a future event with time/place/magnitude; EEW alerts after initiation.

ADVANCED 5 - VOLCANIC MONITORING AND ERUPTION-STYLE UNCERTAINTY

No single signal guarantees an eruption. Monitoring combines:

- earthquake swarms and volcanic tremor;
- deformation from GPS, tilt and satellite radar;
- gas composition and flux;
- thermal anomalies;
- visual, infrasound and satellite ash observations;
- petrology and past deposits.

```
unrest data -> multiple independent indicators
-> scenario probabilities -> alert level
-> exclusion/aviation/community action
```

Viscosity, gas and composition remain central, but conduit geometry, magma recharge, groundwater and dome collapse can shift style. A basaltic system may still produce hazardous fountains, ash and gas, while a silicic system may extrude quietly for a time before explosive transition.

Optional conclusion: Volcano forecasting is scenario-based risk communication under uncertainty, not a declaration that a particular eruption is metaphysically inevitable.

CONSOLIDATED REGISTER NOTES

1. THE COMPLETE EARTHQUAKE MECHANISM

- **Earthquake:** transient ground vibration from sudden energy release, usually fault rupture.
- **Driver versus mechanism:** plate motion loads the region; rupture/slip is the immediate source.
- **Elastic rebound:** load -> lock -> strain -> rupture -> slip -> partial recovery.
- **Stress-fault pairs:** tension-normal; compression-reverse/thrust; shear-strike-slip.
- **Source classes:** tectonic, volcanic, collapse, explosion, induced/reservoir-associated.
- **Depth:** shallow 0-70 km; intermediate 70-300 km; deep 300-700 km.
- **Sequence:** foreshock/mainshock/aftershock labels are relational; swarm lacks one clearly dominant event.

2. SOURCE GEOMETRY AND SEISMOGRAM

- **Focus/hypocentre:** subsurface rupture origin; **epicentre:** vertical surface projection.
- One station's **P-S interval** gives distance; three or more station-distance circles estimate the epicentre.
- Large events rupture finite fault areas; the focus-point model is a simplification.
- Epicentre ≠ worst-damaged locality.

```
seismogram: P first -> S second -> surface waves later
```

3. WAVES AND EARTH INTERIOR

Wave	Motion	Medium	Use
P	Compressional, parallel	Solid + liquid	First arrival; core refraction
S	Shear, perpendicular	Solid only	Liquid outer-core evidence
Love	Horizontal surface shear	Surface	Strong lateral motion
Rayleigh	Elliptical rolling	Surface	Vertical + horizontal motion

- Approximate **P shadow:** 103°-142°.
- Direct **S shadow:** beyond about 103°.
- Do not say the entire core is liquid; outer core liquid, inner core solid by combined evidence.

4. MAGNITUDE, INTENSITY AND DAMAGE

- **Magnitude:** source size; modern large-event use **Mw**.
- **Seismic moment:** rigidity × fault area × average slip.
- **Local-magnitude intuition:** +1 ≈ 10× amplitude and ~31.6× energy.
- **Intensity:** place-specific effects; MMI/MSK-type scale.
- One magnitude; multiple intensities.
- Damage chain: **source × path × site × exposure × vulnerability ÷ capacity**.

5. GLOBAL DISTRIBUTION

- **Circum-Pacific:** trenches, Benioff zones, arcs, megathrusts, tsunamis.
- **Mediterranean-Himalayan:** complex convergence/collision from Mediterranean to Myanmar.
- **Mid-ocean ridges/transforms:** shallow seismicity; ridge basaltic volcanism.
- **Intraplate:** inherited structure reactivation; relatively stable ≠ aseismic.

6. CASCADING EARTHQUAKE HAZARDS

- **Amplification:** soft sediment/basin may enlarge or prolong selected motions.
- **Liquefaction:** loose saturated granular sediment loses effective strength.
- **Lateral spreading:** liquefied soil moves towards free face/down slope.
- **Landslide/rockfall:** marginal slopes fail.
- **Fire/lifeline failure:** utility and access networks cascade.
- **Surface rupture:** siting issue across fault trace.
- **Dam/embankment distress:** shaking + settlement + slope failure.

7. TSUNAMI SPINE

```
submarine megathrust
-> rapid vertical sea-floor displacement
-> long low wave in deep ocean
-> shoaling near coast
-> run-up + inundation + return flow
```

- Not a tide or wind wave.
- First wave need not be largest; withdrawal need not precede every event.
- Source belts: Ring of Fire and Sunda-Andaman.
- Coastal controls: bathymetry, bays, shelf, slope, settlement, evacuation.
- India case: 26 Dec 2004 -> INCOIS/ITEWC warning architecture.

8. VOLCANO ANATOMY AND MAGMA CONTROL

- **Magma** below; **lava** after eruption.
- Chamber -> conduit -> vent -> crater; side vents/dykes/sills.
- **Caldera** = large collapse depression after major withdrawal; not just big crater.
- Silica ↑ -> polymerisation/viscosity ↑ -> degassing harder.
- Temperature ↑ -> viscosity generally ↓.
- Retained volatiles + decompression -> bubbles -> fragmentation/explosion.

9. INTRUSIVE/EXTRUSIVE FORMS AND PRODUCTS

- Intrusive: batholith, laccolith, lopolith, phacolith, dyke, sill.
- Dyke cuts layers; sill parallels layers.
- Products: lava, gases, ash/dust, lapilli, bombs/blocks, density currents, lahars.
- **2024 Prelims official key:** pyroclastic debris + ash/dust + nitrogen compounds + sulphur compounds = all four.

10. ERUPTION STYLES AND VOLCANO TYPES

- Effusive -> explosive: Icelandic/fissure -> Hawaiian -> Strombolian -> Vulcanian -> Plinian/Pelean.
- Shield = fluid flows, gentle; composite = alternating lava/tephra, steep; cinder = fragments; dome = viscous; fissure = plateau.
- Active/dormant/extinct = capability/status evidence, not mandatory life cycle.
- Barren Island = active Indian volcano; 2018 PYQ answer inferred **1 only**.

11. VOLCANIC DISTRIBUTION AND PLUMES

- Divergence: decompression melting -> basalt/ridge/rift.
- Subduction: slab volatiles -> mantle-wedge flux melting -> arc.
- Ring of Fire = trench-slab-earthquake-arc system.
- Plume: head -> flood basalt; tail -> hot spot; moving plate -> age chain.
- Deccan = flood-basalt plateau, not giant cone.
- Plume hypothesis is useful but qualified.

12. VOLCANIC IMPACTS

- Hazards: lava, ash, density currents, lahars, gases.
- Climate: stratospheric sulphate aerosol -> temporary cooling.
- Benefits: soil material, new crust/land, geothermal energy, minerals, tourism.
- Regional impact depends on wind, relief, drainage, island size and exposure.
- 2021 cases: La Soufrière, Nyiragongo, Cumbre Vieja, Semeru.

13. INDIA REGIONAL MAP RECALL

```
Himalaya: collision/thrust + slope cascade
NE/Indo-Burmese: complex convergence/strike-slip
Andaman: subduction + tsunami + Barren volcanism
Kachchh: inherited rift/fault reactivation; Bhuj
Peninsula: relatively stable, not aseismic; Koyna/Latur
Indo-Gangetic: foreland alluvium + amplification/exposure
```

- Case bank: Bhuj 2001; Indian Ocean tsunami 2004; Sikkim 2011; Gorkha/Nepal effects in India 2015.
- Current anchor: NCS/PIB June-July 2026 Kangra-Chamba monitoring.

14. ZONATION AND THE OFFICIAL-STATUS CAUTION

- Macrozonation: broad national design hazard; not prediction or neighbourhood map.
- Familiar publicly verifiable BIS baseline: IS 1893 (Part 1):2016, Zones II-V.
- PIB 17 Dec 2025: official reply refers to Himalayan Zone VI updated map.
- PIB 22 Jul 2026: official reply calls Kangra-Chamba Zone V, highest.
- Exam-safe: cite source/date; do not silently claim settled BIS Zone VI without adopted-standard verification.
- Microzonation: geology + soil + ground motion + groundwater + liquefaction + GIS.
- NCS public page: Delhi 1:10,000 and other city studies.

15. INSTITUTIONS AND RISK REDUCTION

- NCS/MoES: monitoring, event reports, hazard/microzonation.
- GSI: geological and seismotectonic evidence.
- BIS: design/construction/evaluation standards.
- NDMA/NIDM: guidance, preparedness and training.
- States/UTs/ULBs: adoption, building approval, inspection, enforcement.
- INCOIS: tsunami warning.
- Building spine: safe site -> regular form/load path -> ductility/connections -> anchored non-structural elements -> quality control -> retrofit.
- EEW detects after rupture; prediction claims before rupture.
- Household/community: Drop-Cover-Hold On, secure heavy objects, avoid lifts, drills, routes, utility/fire checks.

16. EXAMINER TRAPS

1. Focus ≠ epicentre ≠ worst damage.
2. Magnitude ≠ intensity ≠ loss.
3. S waves do not cross liquids; P waves do.
4. Surface waves are not the only possible damaging waves.
5. “Stable shield” ≠ aseismic.
6. Tsunami ≠ tidal wave.
7. Strike-slip earthquake does not automatically create basin-wide tsunami.
8. Shield/composite = form; Hawaiian/Plinian = style; active/dormant/extinct = status.
9. Caldera ≠ ordinary crater.
10. Lahar ≠ lava; density current ≠ ash fall.
11. Deccan = fissure flood basalt, not cone chain.
12. Zone map ≠ prediction; microzonation ≠ event forecast.
13. Code availability ≠ enforcement.
14. Recent tremor/eruption ≠ proof of deterministic future event.

17. READY INTRODUCTIONS AND CONCLUSIONS

Earthquake introduction: “An earthquake is the sudden release of stored elastic strain, but its disaster outcome is produced by the interaction of source, path, site, exposure and vulnerability.”

Volcano introduction: “Vulcanism transfers heat and material from the Earth's interior to the crust and surface, producing both high-intensity hazards and long-term land, soil and resource benefits.”

India seismic conclusion: “India needs a layered safety system-national hazard maps, city microzonation, code-compliant construction, retrofitted lifelines and rehearsed community response-because prediction cannot substitute for resilience.”

Balanced volcano conclusion: “The developmental value of volcanic landscapes is sustainable only when monitoring and land-use discipline prevent persistent resource benefits from becoming an exposure trap.”

18. ANSWER-WRITING SPINES

10-mark process answer

```
definition -> labelled diagram -> 3-step mechanism -> 2 examples
-> one limitation/trap -> direct conclusion
```

15-mark India risk answer

```
tectonic regions -> ground/site modifiers -> exposure/vulnerability
-> cases -> zonation/microzonation -> qualified policy conclusion
```

20-mark critical answer

```
thesis -> value -> limitations -> official/source inconsistency
-> institutional implementation chain -> graded verdict
```

COMPLETE TOPIC ASCII MASTER FLOW DIAGRAM

High-yield use: Follow the panels as one topic-specific conceptual atlas: root question, classifications, processes or arguments, comparisons, objections and a final answer-writing spine. This text-native master is separate from both graphical flowchart artifacts.

ASCII MASTER FLOW - PANEL 1/8: Geography 03 - Vulcanism and Earthquakes / India Seismic Zones: central question and…

```
GEOGRAPHY 03 — VULCANISM AND EARTHQUAKES / INDIA SEISMIC ZONES — CONCEPTUAL ATLAS


KEY TERMS: endogenic processes • lithospheric plates • convergent margin • divergent margin
KEY TERMS: • transform margin • subduction
                         CENTRAL QUESTION / ROOT
        How should the complete structure of Geography 03 — Vulcanism and Earthquakes / India…
        be defined, related, compared and evaluated?
                                  │
                                  ▼

                       Plate tectonics explains endogenic belts through interactions among…
                                  │
          ┌───────────────────────┼───────────────────────┐
          ▼                       ▼                       ▼
AXIS 1: ENDOGENIC PROCESSES AND THE PLATE-TECTONIC FRAME
  • endogenic processes
  • Mantle heat and gravity drive plate motion, while collision…
AXIS 2: REFRACTION, DISCONTINUITIES AND SHADOW ZONES
  • Refraction
  • The reappearance of P phases beyond the shadow zone results from…
AXIS 3: INTRUSIVE AND EXTRUSIVE FORMS
  • Intrusive
  • Batholiths are very large discordant plutonic bodies later…
          └───────────────────────┼───────────────────────┘
                                  ▼

                         CROSS-PANEL CONTROL
  endogenic processes • lithospheric plates • convergent margin • divergent margin
                                  │
                                  ▼

          Definition → relation → mechanism → comparison → verdict
```

ASCII MASTER FLOW - PANEL 2/8: Plate stress, elastic rebound, faulting and earthquake terminology

```
PLATE STRESS, ELASTIC REBOUND, FAULTING AND EARTHQUAKE TERMINOLOGY
│

KEY TERMS: endogenic processes • lithospheric plates • convergent margin • divergent margin
KEY TERMS: • transform margin • subduction
├── KEY CONCEPTS
│   ├── endogenic processes
│   ├── lithospheric plates
│   ├── convergent margin
│   ├── divergent margin
│   └── transform margin
│
├── ENDOGENIC PROCESSES AND THE PLATE-TECTONIC FRAME
│   ├ TERMS: Plate tectonics explains endogenic belts through interactions among…
│   ├ LOGIC: Mantle heat and gravity drive plate motion, while collision…
│   └ RESULT: Different boundary mechanics produce distinct belts of earthquakes…
├── STRESS, STRAIN, ELASTIC REBOUND AND FAULTS
│   ├ TERMS: Elastic strain is recoverable until failure, whereas permanent…
│   ├ LOGIC: Fault maps identify where rupture has occurred or may be…
│   └ RESULT: Elastic strain is recoverable until failure, whereas permanent…
├── EARTHQUAKE TERMINOLOGY, CAUSES AND CLASSIFICATIONS
│   ├ TERMS: Intermediate and deep earthquakes are strongly associated with…
│   ├ LOGIC: It does not mean every reservoir causes a damaging earthquake
│   └ RESULT: most damaging earthquakes are shallow because the source is closer…
├── FOCUS, EPICENTRE, SEISMOGRAPHS AND LOCATING AN EVENT
│   ├ TERMS: Modern location uses many stations, a velocity model and iterative…
│   ├ LOGIC: The earliest small motion is normally the P arrival, followed by S…
│   └ RESULT: Three-circle triangulation is the learner model, not the full…
│
└── CONVERGENCE
    ├ Distinguish the branches before combining them
    └ Reconnect each branch to the topic's central explanatory problem
```

ASCII MASTER FLOW - PANEL 3/8: Seismic waves, shadow zones, magnitude, intensity and distribution

```
SEISMIC WAVES, SHADOW ZONES, MAGNITUDE, INTENSITY AND DISTRIBUTION


KEY TERMS: P • S • Love • Rayleigh Waves • To-and-fro parallel to travel direction • Solids
KEY TERMS: and liquids
CAUSE / CONDITION 1: “Surface waves are most destructive” is a useful rule, not an…
          │
          ├── MECHANISM
          │   S waves require shear rigidity and therefore do not pass through a…
          │
          └── EFFECT / CONTRAST
              P waves can pass through both solid and liquid but change velocity…
          │
          ▼
CAUSE / CONDITION 2: The reappearance of P phases beyond the shadow zone results…
          │
          ├── MECHANISM
          │   The reappearance of P phases beyond the shadow zone results from…
          │
          └── EFFECT / CONTRAST
              This is not evidence that the whole core is liquid
          │
          ▼
CAUSE / CONDITION 3: Intensity is not a synonym for economic loss because two…
          │
          ├── MECHANISM
          │   Mw converts this physical source measure to a logarithmic magnitude
          │
          └── EFFECT / CONTRAST
              Intensity is not a synonym for economic loss because two buildings…
          │
          ▼
CAUSE / CONDITION 4: Ridge earthquakes are shallow because the lithosphere is thin…
          │
          ├── MECHANISM
          │   Ridge earthquakes are shallow because the lithosphere is thin and…
          │
          └── EFFECT / CONTRAST
              Bhuj, Latur and New Madrid show that plate interiors are…
          │
          ▼
SYSTEM RESULT: causes interact; mechanism explains why the outcome follows.
```

ASCII MASTER FLOW - PANEL 4/8: Secondary hazards, tsunami process and volcanic anatomy/forms

```
SECONDARY HAZARDS, TSUNAMI PROCESS AND VOLCANIC ANATOMY/FORMS


KEY TERMS: Secondary Hazards • Site Response • The Risk Equation • Liquefaction •
KEY TERMS: Floodplains, deltas, reclaimed land… • Lateral spreading
UPSTREAM / OUTER SETTING
        │
        ▼
┌───────────────────────────────────────────────────────────────────────┐
│ ZONE / LAYER A                                                        │
│ SECONDARY HAZARDS, SITE RESPONSE AND THE RISK EQUATION                │
│ Process: Urban disaster risk is often a network problem: a…           │
│ Result : Liquefaction is not “soil melting.” It is a temporary loss…  │
└───────────────────────────────────────────────────────────────────────┘
        │
        ▼
┌───────────────────────────────────────────────────────────────────────┐
│ ZONE / LAYER B                                                        │
│ TSUNAMI GENERATION, PROPAGATION AND COASTAL IMPACT                    │
│ Process: “Move to 10 metres” is not a universal rule                  │
│ Result : In deep water, a tsunami can pass beneath ships with…        │
└───────────────────────────────────────────────────────────────────────┘
        │
        ▼
┌───────────────────────────────────────────────────────────────────────┐
│ ZONE / LAYER C                                                        │
│ VOLCANO ANATOMY, MAGMA, LAVA, SILICA, GAS AND VISCOSITY               │
│ Process: Dissolved volatiles exsolve into bubbles                     │
│ Result : Basaltic magma is generally hotter, less silica-rich and…    │
└───────────────────────────────────────────────────────────────────────┘
        │
        ▼
SPATIAL CONTROL: locate the process, then compare scale and regional expression.
```

ASCII MASTER FLOW - PANEL 5/8: Eruption styles, volcano classes, plate settings and hot spots

```
ERUPTION STYLES, VOLCANO CLASSES, PLATE SETTINGS AND HOT SPOTS


KEY TERMS: Intrusive • Extrusive Forms • Volcanic Products • D: all four • Lava flow •
KEY TERMS: Molten rock at surface
```

MODEL / BRANCH	CORE POSITION	DISTINCTION / RESULT
INTRUSIVE AND EXTRUSIVE FORMS	Product classification is also a risk map: ballistic blocks…	UPSC Prelims 2024 asked whether pyroclastic debris, ash and dust…
ERUPTION STYLES, VOLCANO TYPES AND ACTIVITY STATUS	A volcano with no historical eruption may still be capable of…	“Active, dormant and extinct” are descriptive status classes, not a…
PLATE-SETTING VOLCANISM, RING OF FIRE AND MID-OCEAN RIDGES	The Ring of Fire includes trenches, island arcs, continental arcs…	The Mediterranean–Himalayan belt is not one uniform volcanic arc
MANTLE PLUMES, HOT SPOTS AND FLOOD BASALTS	Flood basalts erupt mainly from fissures and build plateaux through…	Hot spots can supply a reference frame for plate direction and…

```
                                    │
                                    ▼
COMPARE ON THE SAME AXIS: definition • mechanism • consequence • limit
```

ASCII MASTER FLOW - PANEL 6/8: India seismic zones, microzonation, resilience and warning limits

```
INDIA SEISMIC ZONES, MICROZONATION, RESILIENCE AND WARNING LIMITS


KEY TERMS: Constructive • Destructive • Climatic • Environmental Effects • Lava • Burial,
KEY TERMS: burning, cutting access
CAUSE / CONDITION 1: The climatic effect depends on what reaches the stratosphere…
          │
          ├── MECHANISM
          │   Short-lived cooling after a major explosive eruption is principally…
          │
          └── EFFECT / CONTRAST
              Ash can eventually contribute nutrients, but fresh ash may…
          │
          ▼
CAUSE / CONDITION 2: The 2001 Bhuj earthquake demonstrates severe intraplate risk…
          │
          ├── MECHANISM
          │   The 2001 Bhuj earthquake demonstrates severe intraplate risk in…
          │
          └── EFFECT / CONTRAST
              The Peninsular shield is relatively stable, not aseismic
          │
          ▼
CAUSE / CONDITION 3: Microzonation must feed building regulation, land use…
          │
          ├── MECHANISM
          │   Because the corresponding adopted BIS standard/map could not be…
          │
          └── EFFECT / CONTRAST
              Microzonation must feed building regulation, land use, lifeline…
          │
          ▼
CAUSE / CONDITION 4: Institutions, Resilient Construction, Early Warning Limits…
          │
          ├── MECHANISM
          │   It is not prediction
          │
          └── EFFECT / CONTRAST
              The implementation gap is often the decisive risk multiplier: a…
          │
          ▼
SYSTEM RESULT: causes interact; mechanism explains why the outcome follows.
```

ASCII MASTER FLOW - PANEL 7/8: Geography 03 - Vulcanism and Earthquakes / India Seismic Zones: traps, PYQ routes and…

```
GEOGRAPHY 03 — VULCANISM AND EARTHQUAKES / INDIA SEISMIC ZONES: TRAPS, PYQ ROUTES AND…


KEY TERMS: endogenic processes • lithospheric plates • convergent margin • divergent margin
KEY TERMS: • transform margin • subduction
                         EXAM APPLICATION ROOT
                                  │
          ┌───────────────────────┼───────────────────────┐
          ▼                       ▼                       ▼
    CLOSE-OPTION TRAPS         PYQ / QUESTION ROUTES      ANSWER USE
├── Do not treat every convergent belt as volcanically identical…
├── “Richter scale” properly refers to local magnitude developed for a…
├── Narcondam is volcanic but should not be substituted for Barren…
├── retrofitting may add confinement, bracing, shear walls, jacketing…
│
├── Exact wording is reproduced from the routed local UPSC question papers
├── VERIFIED PYQ 1 — UPSC Prelims 2018 GS-I Q85: Barren Island
├── Why this PYQ matters: UPSC combines activity status, map location and…
├── VERIFIED PYQ 2 — UPSC Prelims 2023 GS-I Q65: P and S waves
│
└── Plate tectonics explains why earthquakes, volcanism and mountain…
└── Strike-slip movement with little vertical displacement is less…
└── Institutions, Resilient Construction, Early Warning Limits And…
                                  │
                                  ▼
CONTROL: identify demand → select exact evidence → expose trap → qualify.
```

ASCII MASTER FLOW - PANEL 8/8: Geography 03 - Vulcanism and Earthquakes / India Seismic Zones: integrated revision and…

```
GEOGRAPHY 03 — VULCANISM AND EARTHQUAKES / INDIA SEISMIC ZONES — ANSWER / REVISION SPINE


KEY TERMS: endogenic processes • lithospheric plates • convergent margin • divergent margin
KEY TERMS: • transform margin • subduction
1. DEFINE THE PRECISE TOPIC / TERM
   └─ Plate tectonics explains endogenic belts through interactions among…
                  │
                  ▼

2. MAP THE MAIN TYPES, STAGES OR INSTITUTIONS
   └─ endogenic processes
                  │
                  ▼

3. EXPLAIN THE MECHANISM / ARGUMENT / CAUSAL LINK
   └─ Mw converts this physical source measure to a logarithmic magnitude
                  │
                  ▼

4. COMPARE ON A COMMON ANALYTICAL AXIS
   └─ convergent margin
                  │
                  ▼

5. TEST WITH OBJECTION, LIMIT, EXCEPTION OR TRAP
   └─ retrofitting may add confinement, bracing, shear walls, jacketing…
                  │
                  ▼

6. GIVE A QUALIFIED, TOPIC-SPECIFIC VERDICT
   └─ Institutions, Resilient Construction, Early Warning Limits And…

HIGH-YIELD FORMULA
  ROOT CONCEPT
      + CLASSIFICATION / SEQUENCE
      + MECHANISM / EVIDENCE
      + COMPARISON / APPLICATION
      + OBJECTION / LIMIT
      = QUALIFIED ANSWER
```